

DMS201: Introduction to Management

Module-3: Operations Management

(Production Management)

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Production Management

- **Introduction to production management**
- Capacity planning and process analysis
- Facility location
- Aggregate production planning
- Materials requirement planning
- Sequencing and scheduling

Introduction to Production Management

Introduction to Operations Management

- Operations Management (OM) refers to the business processes involved with planning, organizing and monitoring the production process.
- The essence of OM is to use efficient methods of converting raw materials and labour into goods and services
- The ideas of OM is to optimize the production and delivery process.
- The core ideas of Production and Operations Management are (i) **Inventory Management**, (ii) **Production Planning** and (iii) **Scheduling**

Introduction to Operations Management

- For any business to run **efficiently** and **effectively** one needs to utilize all available resources optimally to meet customers' requirements.
- In doing so the business needs to **design, control** and **manage** the production of goods and services in the right manner.

Introduction to Operations Management (Key Points)

- **Manufacturing**: technology, advanced materials.
- **Strategic dimensions**: quality, cost, reliability, speed, flexibility.
- **Global competition**: balance of trade, market share, job creation, cost of production, cost of labour.

Introduction to Operations Management (Key Points)

- **Strategic initiatives:** Business Process Engineering (BPE), Just In Time (JIT), KANBAN, Lean Manufacturing, KAIZEN, Agile manufacturing.
- **Product and process life cycles:** job shop production, continuous production.
- **Learning and experience curves.**
- **Capacity growth planning.**

Introduction to Operations Management (Sectors and Companies)

■ Aerospace industries

- ❖ RTX (Raytheon Technologies) of USA, Revenue US\$ 80.7 billion in 2024, <https://www.rtx.com/>
- ❖ Airbus of Europe, Revenue US\$ 74.7 billion in 2024, <https://www.airbus.com/>
- ❖ Lockheed Martin of USA, Revenue US\$ 71.0 billion in 2024, <https://lockheedmartin.com/>
- ❖ Boeing of USA, Revenue US\$ 66.5 billion in 2024, <https://www.boeing.com/>
- ❖ GE Aerospace of USA, Revenue US\$ 38.7 billion in 2024, <https://www.geaerospace.com/>

Introduction to Operations Management (Sectors and Companies)

■ Chemical industries

- ❖ BASF of Germany, Revenue US\$ 70.6 billion in 2024, <https://www.basf.com>.
- ❖ Sinopec of China, Revenue US\$ 58.1 billion in 2024, <http://www.sinopecgroup.com/>.
- ❖ Dow of USA, Revenue US\$ 43.0 billion in 2024, <https://dow.com/>.
- ❖ PetroChina of China, Revenue US\$ 42.8 billion in 2024, <http://www.petrochina.com.cn>.
- ❖ ExxonMobil of USA, Revenue US\$ 41.1 billion in 2024, <https://corporate.exxonmobil.com/>.

Introduction to Operations Management (Sectors and Companies)

■ Computer/Electronics industries

- ❖ Amazon of USA, Revenue US\$ 637.9 billion in 2024, <https://www.amazon.com/>.
- ❖ Apple Inc. of USA, Revenue US\$ 416.0 billion in 2024, <https://www.apple.com/>.
- ❖ Alphabet Inc. of USA, Revenue US\$ 350.0 billion in 2024, <https://abc.xyz/>.
- ❖ Microsoft of USA, Revenue US\$ 281.7 billion in 2024, <https://www.microsoft.com/>.
- ❖ Samsung Electronics of South Korea, Revenue US\$ 220.3 billion in 2024, <https://www.samsung.com/>.

Introduction to Operations Management (Sectors and Companies)

■ Agriculture/Food industries

- ❖ Cargill of USA, Revenue US\$ 160.0 billion in 2024, <http://www.cargill.com/>.
- ❖ Archer Daniels Midland (ADM) of USA, Revenue US\$ 85.5 billion in 2024, <http://adm.com/>.
- ❖ Syngenta Group of China, Revenue US\$ 28.8 billion in 2024, <https://www.syngenta.com/>.
- ❖ Bayer Crop Science of Germany, Revenue US\$ 26.6 billion in 2024, <https://www.bayer.com/>.
- ❖ Corteva Agriscience of USA, Revenue US\$ 16.9 billion in 2024, <http://corteva.com/>.

Introduction to Operations Management (**Sectors and Companies**)

■ **Machinery** industries

- ❖Caterpillar Inc. of USA, Revenue US\$ 64.8 billion in 2024,
<https://www.caterpillar.com/>.
- ❖John Deere of USA, Revenue US\$ 51.7 billion in 2024,
<https://www.deere.com>.
- ❖Komatsu Ltd.of Japan, Revenue US\$ 26.0 billion in 2024,
<http://komatsu.jp/en>.
- ❖XCMG Group of China, Revenue US\$ 12.6 billion in 2024,
<http://en.xcmg.com/en-ap/>.
- ❖Sany Heavy Industry of China, Revenue US\$ 10.9 billion in 2024,
<https://www.sanyglobal.com/>.

Introduction to Operations Management (Sectors and Companies)

- **Motor vehicles/Transportation** industries
 - ❖ Toyota Motor Corp. of Japan, Revenue US\$ 410.9 billion in 2024, <https://global.toyota/>.
 - ❖ Volkswagen Group of Germany, Revenue US\$ 350.0 billion in 2024, <https://www.volkswagen-group.com/>.
 - ❖ General Motors of USA, Revenue US\$ 187.4 billion in 2024, <https://www.gm.com/>.
 - ❖ Ford Motor Company of USA, Revenue US\$ 185.0 billion in 2024, <https://www.ford.com/>.
 - ❖ Mercedes-Benz Group of Germany, Revenue US\$ 151.9 billion in 2024, <https://group.mercedes-benz.com/>.

Introduction to Operations Management (Sectors and Companies)

- **Pharmaceutical/Medicine** industries
 - ❖ Johnson & Johnson of USA, Revenue US\$ 88.8 billion in 2024, <https://www.jnj.com/>.
 - ❖ Roche of Switzerland, Revenue US\$ 69.2 billion in 2024, <https://www.roche.com/>.
 - ❖ Merck & Co. of USA, Revenue US\$ 64.2 billion in 2024, <https://www.msd.com/>.
 - ❖ Pfizer of USA, Revenue US\$ 63.2 billion in 2024, <https://www.pfizer.com/>.
 - ❖ AbbVie of USA, Revenue US\$ 56.3 billion in 2024, <https://www.abbvie.com/>.

Introduction to Operations Management (Sectors and Companies)

- **Financial Services** industries
 - ❖ Transamerica Corporation of USA, Revenue US\$ 245.5 billion in 2024, <https://www.transamerica.com/>.
 - ❖ Ping An Insurance Group of China, Revenue US\$ 191.5 billion in 2024, <https://group.pingan.com/>.
 - ❖ Industrial and Commercial Bank of China of China, Revenue US\$ 182.8 billion in 2024, <https://www.icbc.com.cn/>.
 - ❖ China Construction Bank of China, Revenue US\$ 172.0 billion in 2024, <https://en.ccb.com/>.
 - ❖ Agricultural Bank of China of China, Revenue US\$ 153.9 billion in 2024, <https://www.abchina.com/en/>.

Introduction to Operations Management (Sectors and Companies)

■ **Steel/Metal** industries

- ❖ Glencore of Switzerland, Revenue US\$ 230.9 billion in 2024, <https://www.glencore.com/>.
- ❖ China Minmetals of China, Revenue US\$ 115.8 billion in 2024, <http://www.minmetals.com/>.
- ❖ ArcelorMittal of Luxembourg, Revenue US\$ 62.4 billion in 2024, <http://corporate.arcelormittal.com/>.
- ❖ China Baowu Group of China, Revenue US\$ 44.7 billion in 2024, <https://www.baowugroup.com/en/>.
- ❖ Nippon Steel Corporation of Japan, Revenue US\$ 58.4 billion in 2024, <https://www.nipponsteel.com/>.

Introduction to Operations Management

(Historical Development)

Concepts/Ideas	Time	Essence
Industrial Revolution	1700s	Using machine power to do human work
Scientific Management	Early 1900s	Use of scientific techniques
Human Relationship	1930 – 1960	Consideration of Human element
Management Science	1940 – 1960	Development of quantitative techniques
Computer	1960s	Use of computer to handle large amount of data
Environmental Issues	1970s	Waste reduction and minimizing effect on environment
Just in Time (JIT)	1980s	Achieve high volume of production
Total Quality Management (TQM)	1980s	Find and eliminate causes of production defects

Introduction to Operations Management

(**Historical Development**)

Concepts/Ideas	Time	Essence
Reengineering	1980s	Redesigning processes
Global competition	1980s	Competition in world business
Flexibility	1990s	Customization of product for customers
Time based competition	1990s	Time reduction
Supply Chain Management (SCM)	1990s	Flow of materials
E-Commerce	2000s	Internet and WWW
Outsourcing	2000s	Flattening of the world

Introduction to Operations Management

(Historical Development)

- **Industrial Revolution:** Steam engine, interchangeable parts, division of labour (Wealth of Nation: Adam Smith), etc.
- **Scientific Management:** Productivity, Frederick Taylor scientific management techniques, Work-Flow study, Time & Motion study, Standardization, Differential piece wage system, etc.
- **Human Relationship:** Hawthorne study, human & environment relationship study, job engagement, job enlargement, job enrichment, etc.
- **Management Science:** Inventory models, Ideas of using mathematical models for production/inventory, etc.

Introduction to Operations Management

(Historical Development)

- **Computer**: Material Requirement Planning (MRP), Data processing, Forecasting, Scheduling, Inventory control, Project Management, Logistics, etc.
- **Environmental Issues**: Pollution, Health issues, degradation of environment, etc.
- **Just in Time (JIT)**: Continuous improvement, organizational efficiency, coordination of flow of materials, minimum amount of inventory, etc.
- **Total Quality Management (TQM)** : W. Edward Deming, Joseph M. Juran, quality circle, quality control charts, quality improvement, cost of poor quality, Pareto principle, etc.

Introduction to Operations Management

(Historical Development)

- **Reengineering**: Customer focus, rethink from scratch, cross functional work, simplification, removing redundancy, etc.
- **Global competition**: Population increase, competitive advantage, resource constraints, transportation, etc.
- **Flexibility**: Mass customization, Radical product design, High volume of production, lowering of cost of mass production, etc.
- **Time based competition**: Better connectivity, global reach, ease of communication and hence information reaching fast, etc.

Introduction to Operations Management

(Historical Development)

- **Supply Chain Management (SCM)** : Information technology, Logistics, reverse logistics, timely delivery, perishable good transportation, global market, etc.
- **E-Commerce**: Business to Business (B2B), Business to Customer (B2C), Customer to Customer (C2C), etc.
- **Outsourcing**: Lean Manufacturing, Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), Massive investment in technology, Internet, Email, world wide internet connectivity, unrestricted movement of labour, cross functional decision making, etc.

Introduction to Operations Management

A study of the **technology**, **competitiveness**, **history**, etc., shows the following countries have an edge

- Germany: printing presses, luxury cars, chemicals.
- Switzerland: pharmaceuticals, chocolate.
- Sweden: heavy trucks, mining equipment.
- United States: personal computers, software, films.
- Japan: automobiles, consumer electronics, robotics.

Introduction to Operations Management

Technology and **competitiveness** are advantageous based on the following reasons

- Historical
- Macroeconomic factors
- National character
- Natural resources
- Government policies
- Labour/Education
- Management practices

Introduction to Operations Management (Product Life cycle)

■ Start-up

- ❖ Cost for developing in the initial stages can be high.
- ❖ Production can be high or low depending on technology or it being an unique product.
- ❖ Distribution is high depending on the customer base which is being targeted.
- ❖ Competition is low, as no existing/new company has the new product.

Introduction to Operations Management (Product Life cycle)

■ Rapid growth

- ❖ Competition is high, as many competitors enter the market.
- ❖ Different Pricing strategies can be used, based on the customer base and the latter's requirements.
- ❖ Manufacturing process improvement happens as expertise is obtained and innovation takes place.
- ❖ Flexibility and modularization of manufacturing may be needed as mass production is required, plus the fact that cost of production per unit is reduced.

Introduction to Operations Management (Product Life cycle)

■ **Maturation**

- ❖ Maintain/improve brand loyalty as competition increases and different competitors give unique services/products to customers.
- ❖ Pricing becomes competitive, with improvement of technology, expertise in production, better distribution system, advantages of competitors, etc.
- ❖ Cost savings is ensured through production control/distribution.

Introduction to Operations Management

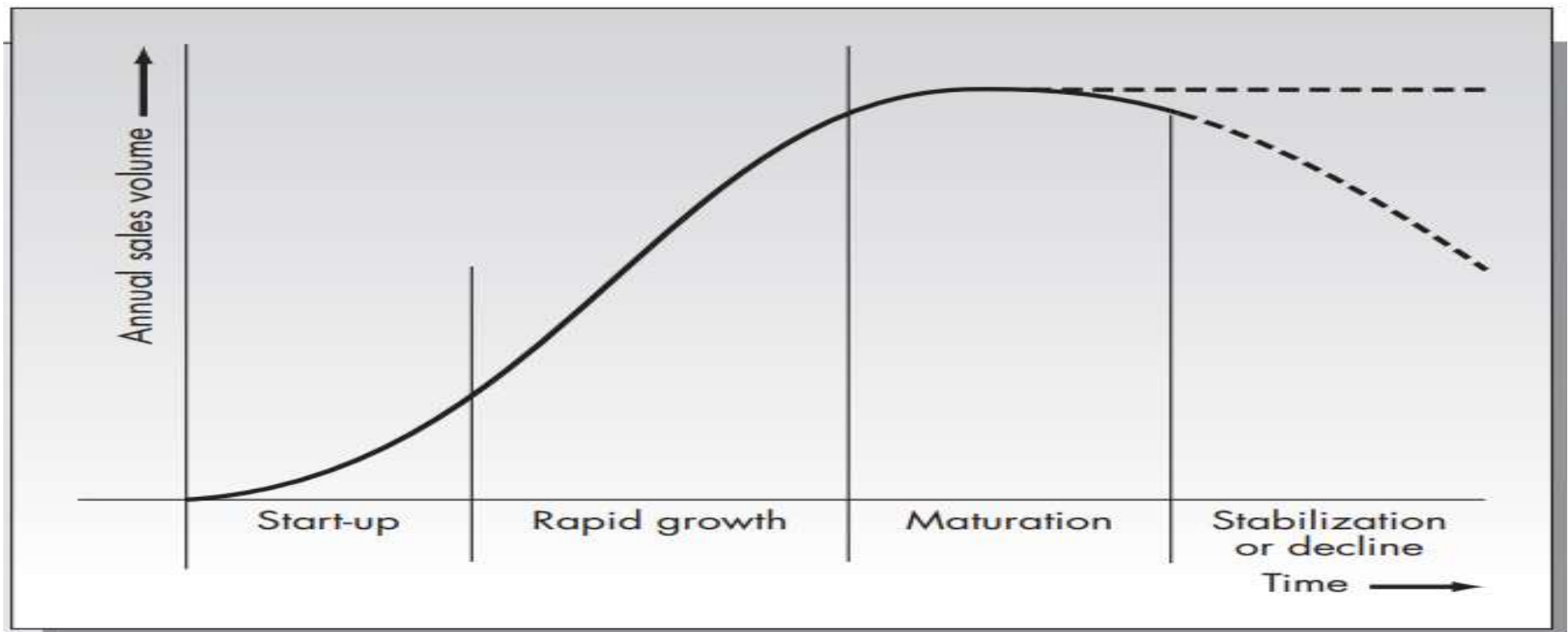
(Product Life cycle)

■ Stabilization and/or decline

- ❖ Obsolescence of product to be overcome, as new and better products enter the market at better prices.
- ❖ Further investments to be minimal, as it may not be prudent at all to invest in new technologies as the product may have outlived its requirement for customers or any further feature or quality characteristics may not attract new or even the old customers.

Introduction to Operations Management

(**Product Life cycle** (Ref: Production and Operations Analysis, Steven Nahmias and Tava Lennon Olsen))



Introduction to Operations Management (Process Life Cycle)

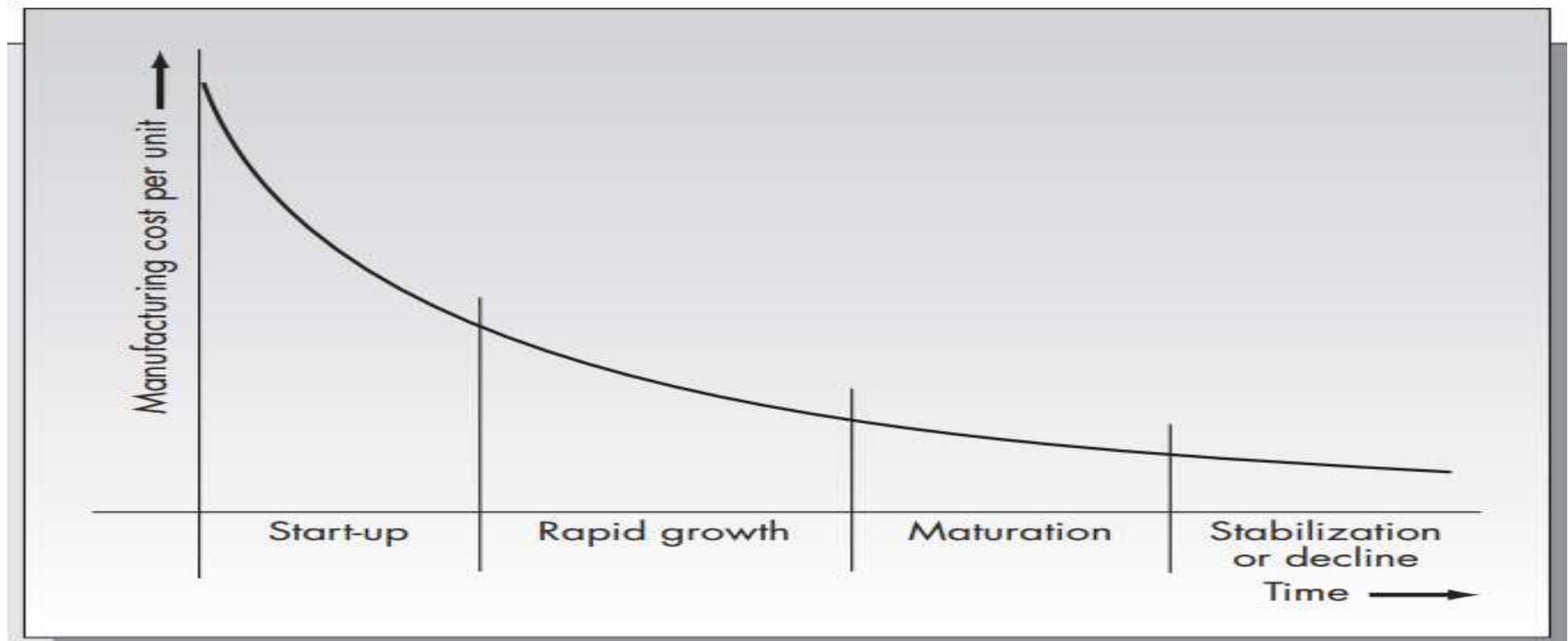
- **Start up**: Job shop production, Volume is low, Selling price is low.
- **Rapid growth**: Innovation sets in, Higher volume, Automation, Batch processing production, Assembly line, Selling price increases, Competition set in.

Introduction to Operations Management (Process Life Cycle)

- **Maturation:** Production process standardized, No innovation, Volume stabilizes, Competition of new better products starts coming.
- **Stabilization/Decline:** Volume decreases, Decrease in prices in order to compete.

Introduction to Operations Management

(**Process Life Cycle** (Ref: Production and Operations Analysis, Steven Nahmias and Tava Lennon Olsen))



Introduction to Operations Management

(Product versus Process Matrix)

Process structure Process life-cycle stage	I: Low Volume, Low Standardization	II: Multiple products, Low Volume	III: Few major products, higher volume	IV: High volume, high standardization, commodity products
I: Job shop	Custom furniture maker	Specialty Machinery Manufacturing	Jewelry making	Machine Shops
II: Batch	R&D Labs	Chain of Bakeries	Automotive Industry	Pharmaceuticals
III: Assembly line	Specialized medical equipment	Customized automotive manufacturing	Smart phones	Fast moving consumer goods
IV: Continuous flow	Custom Machine Fabrication	Custom Chemical Compounds	Steel Manufacturing	Oil Refineries

Introduction to Operations Management (**Learning and Experience Curve**)

Factors which effect are

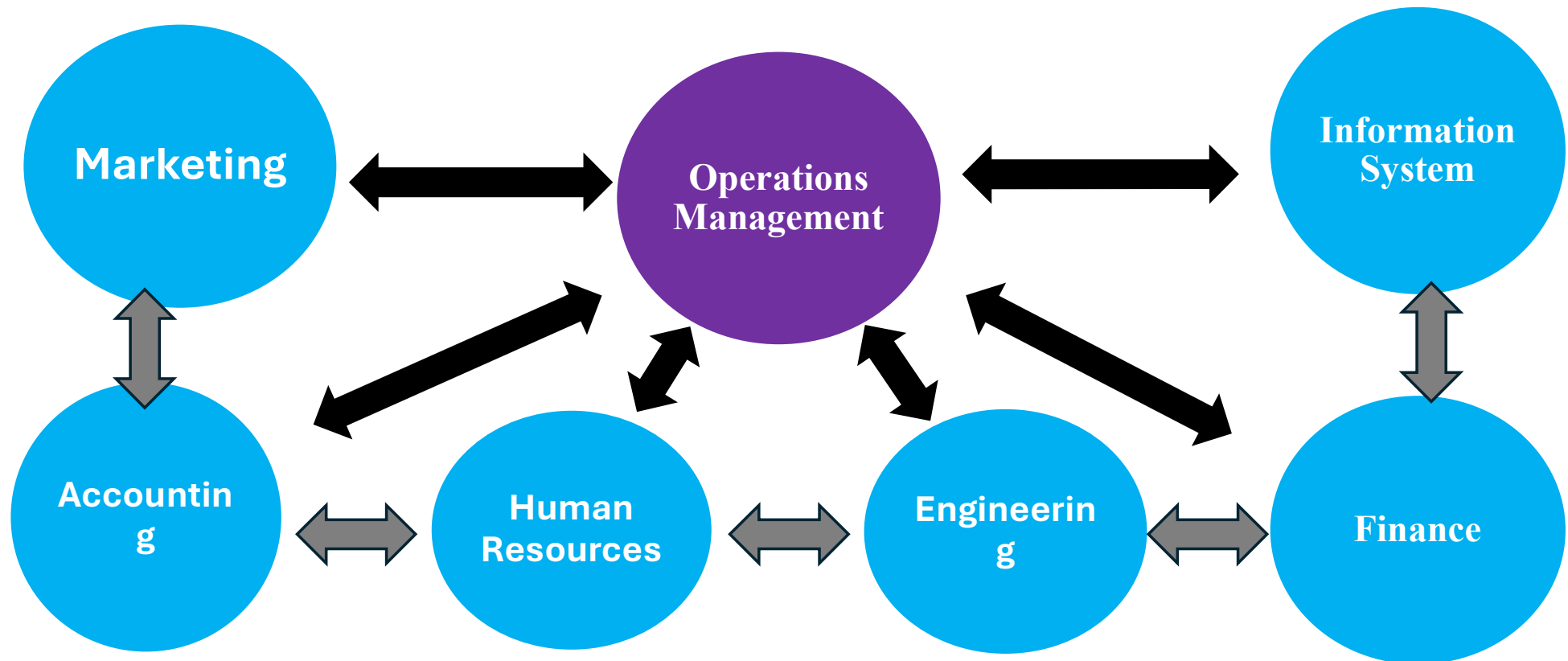
- Improvements in production methods.
- Improvements in the reliability and efficiency of the tools and machines used.
- Better product design.
- Improvements in production scheduling and inventory control systems.
- Better organization of the workplace.

Introduction to Operations Management

(Key questions in Operations Management)

- What is/are the products/services?
- Who are the customers and what is their demands?
- How much should the customers be charged?
- How efficiently/effectively can products/services be delivered?
- When will demand be fulfilled?
- Where will demand be fulfilled?

Introduction to Operations Management (**Information Flow**)

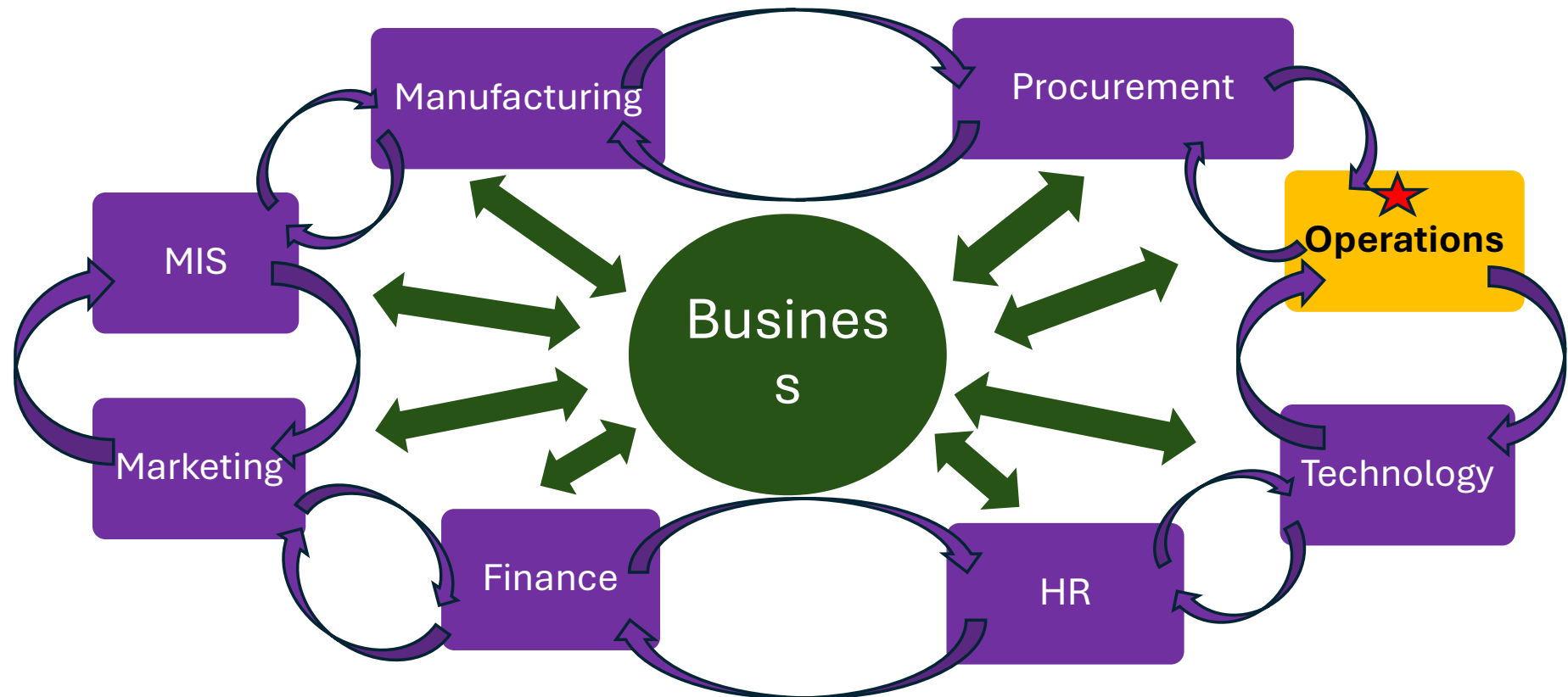


Introduction to Operations Management

(Operations Strategy)

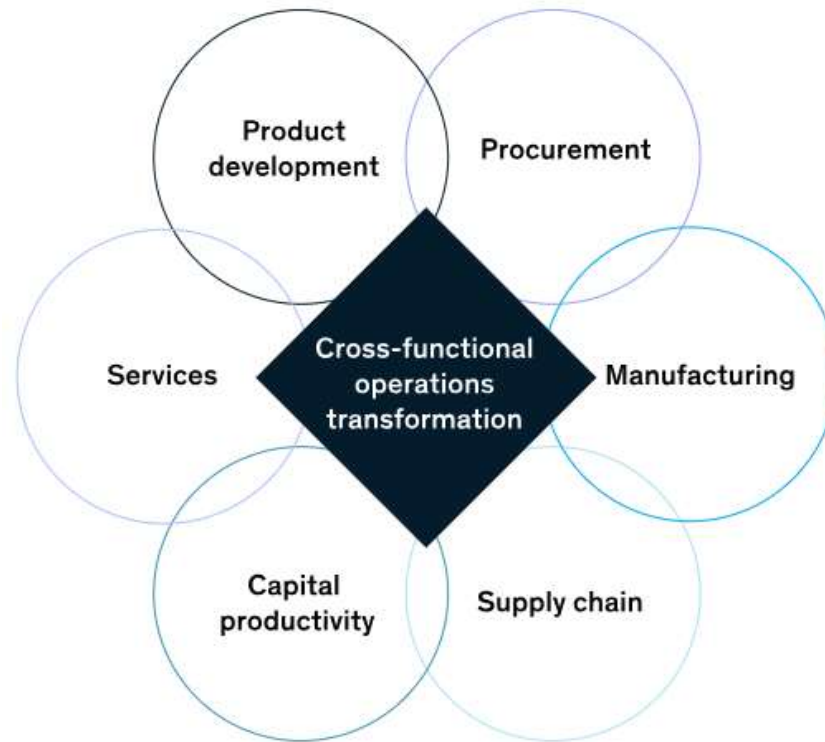
- The role of operations strategy is to provide a plan for the operations function so that it can make the best use of its resources.
- Operations strategy specifies the policies and plans for using the organization's resources to support its long-term competitive strategy.
- Operations strategy must be aligned with the company's business strategy and enable the company to achieve its long-term plan.

Introduction to Operations Management (Operations Strategy)



Introduction to Operations Management

(Operations Strategy: McKinsey & Company)



Introduction to Operations Management (**Competitive Strategy Based on**)

- Cost
- Quality
- Flexibility
- Time
- Technology
- Inherent advantages
- Policies

Introduction to Operations Management

(Competitive Strategy Measurement)

- Productivity = (Output Produced/Input Used)
- Partial Productivity
 - ❖ (Output/Labour)
 - ❖ (Output/Machines)
 - ❖ (Output/Material)
 - ❖ (Output/Capital)
- Multifactor Productivity
 - ❖ {Output/(Labour + Machines)}
 - ❖ {Output/(Labour + Materials)}
 - ❖ {Output/(Labour + Capital + Energy)}

Introduction to Operations Management

(Example # 01)

- Two workers paint tables in a furniture shop. If the workers paint 22 tables in 8 hours. It is also given that the value of output is Rs. 3,82,00/- while labour and electricity costs are Rs. 16,800/- and Rs. 9,800/- respectively.
- Labour productivity = $22 \text{ tables} / (2 \text{ workers} \times 8 \text{ hours}) = 1.375 \text{ tables/hour}$.
- Multi-factor productivity = $38200 / (16800 + 9800) = 1.436$

Introduction to Operations Management (**Product Design**)

- Customers, Surveys & Questionnaires, User Interviews & Focus Groups, Usability Testing, Online Reviews & Social Media, Design experts, Cross functional team members, Product analyst, Feedback Widgets/Forms
- Competitors, Industry experts, Suppliers/Vendors, Purchasers, Retailers
- Bench marking, Reverse engineering

Introduction to Operations Management

(Technology Affecting Production and Operations Management)

- **Enterprise Resource Planning (ERP):** Hardware/Software which integrates and automates the business processes (finance, operations management, manufacturing, sales, procurement, marketing, human resource management, management information processing system, etc.)
- **Global Positioning Systems (GPS):** Satellite based navigational system for precise location/coordinates.

Introduction to Operations Management

(Technology Affecting Production and Operations Management)

- **Radio Frequency Identification (RFID)**: Wireless technology used for ease of tracking and location of objects, etc. RFID is a part and parcel of Automatic Identification and Data Capture (which in general includes Quick Response (QR) Codes, Bar Codes, Magnetic Stripe, Biometric, Smart Card, Voice Recognition, etc.).
- **Automated Material Handling System**: Use of technology for handling materials for movement, etc. Ideas conveyed under this concept can be related to Automated storage and retrieval system, Bulk material handling, Conveyor system, Industrial robot, Material handling equipment, Warehouse design/layout.

Introduction to Operations Management

(Technology Affecting Production and Operations Management)

- **Flexible Manufacturing System (FMS):** Automated set up for production using robotics, machines, material handling equipment, etc. Ideas of Agile manufacturing, Lean manufacturing come under FMS.
- **Robotics, Numerical Control (NC) machines, Computer Numerical Control (CNC) machines:** Robotics are used for multi-tasking for loading/unloading, designing, production, etc., of products.

Introduction to Operations Management

(Technology Affecting Production and Operations Management)

- **Computer-Aided Design (CAD)**: Computers or workstations used to aid in the creation, modification, analysis, or optimization of a design. Other related ideas are: **Computer-aided Engineering (CAE)** and **Finite Element Analysis (FEA)**, **Finite Element Method (FEM)**, etc.
- **Computer-Integrated Manufacturing (CIM)**: Computers used to manage & control the entire manufacturing & production process.

Introduction to Operations Management (Inventory/Stock)

- Inventory/Stock is the quantity of the goods/materials held for production/utilization/sale.
- Reasons for having inventory/stock are: time to order, seasonality, emergency, uncertainty, economic of scale, appreciation of value, to compete against competitor, etc.

Introduction to Operations Management

(Inventory/Stock: Important terms/ideas/concepts)

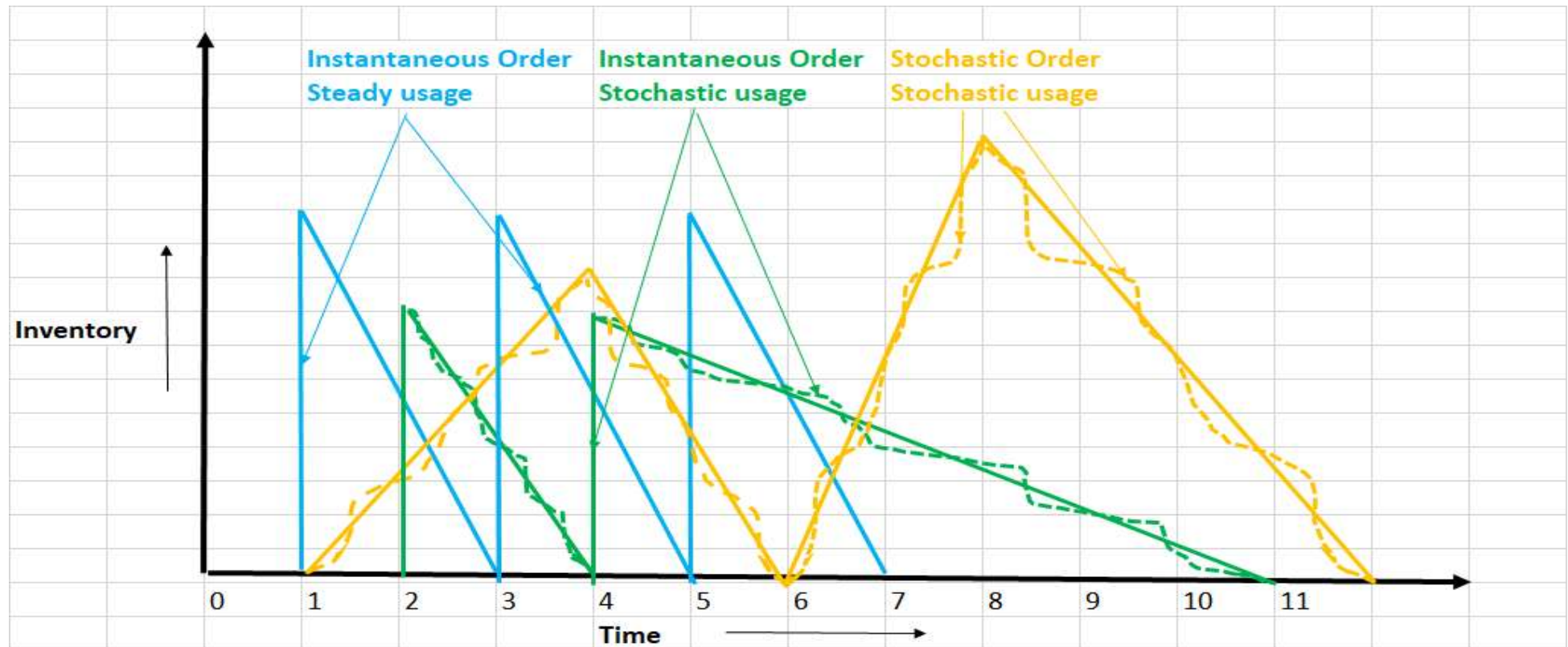
- ABC Analysis
- First In First Out (FIFO), Last In First Out (LIFO), etc.
- Safety Stock
- Reorder point
- Carrying Cost
- Stock out
- Back Ordering
- Lead time
- Inventory Control
- Inventory Optimization
- Economic Order Quantity (EOQ)
- Inventory Turnover
- Just In Time (JIT)
- KANBAN
- Dead Stock

Introduction to Operations Management

(Inventory/Stock: Associated Costs)

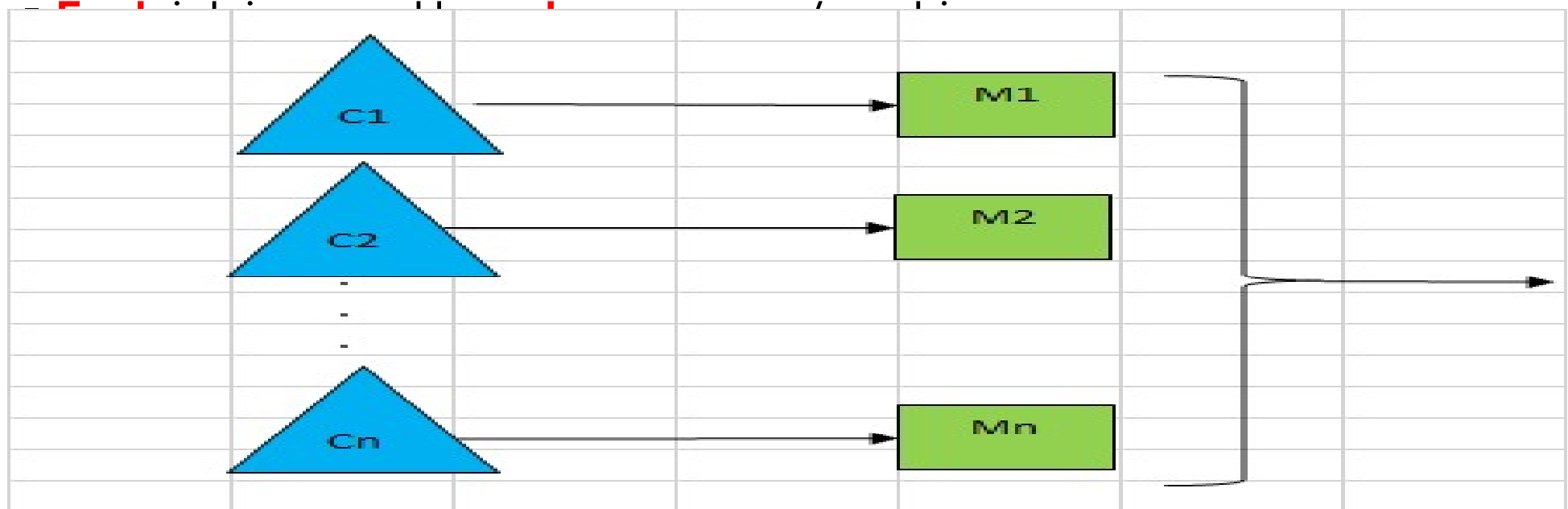
- Processing/Ordering
- Purchase
- Capital/Set up
- Transportation
- Inspection
- Handling
- Warehouse/Holding
- Back ordering
- Shortage/Lost sale/Stock out
- Insurance
- Taxes
- Obsolescence
- Spoilage/Wastage/Damage
- Salary/Wages

Introduction to Operations Management (Inventory/Stock)



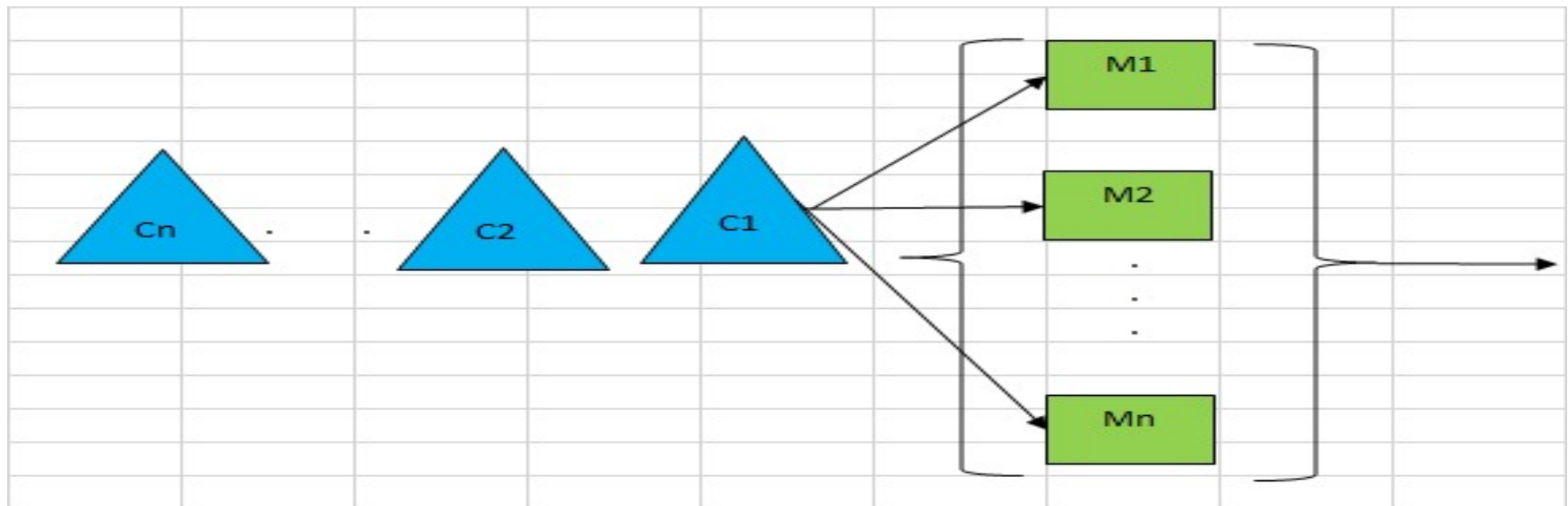
Introduction to Operations Management (Flow Process Diagram)

- n **parallel** processes, each with its **own/individual** dedicated **waiting line**.



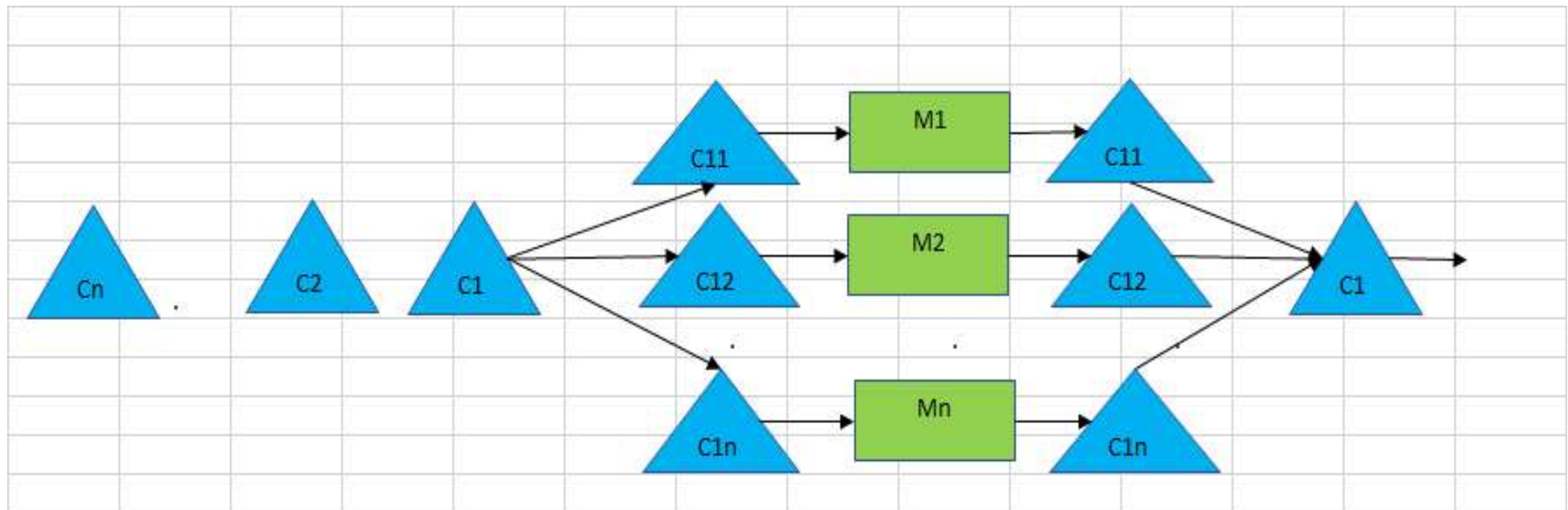
Introduction to Operations Management (Flow Process Diagram)

- n **serial/sequence** process, with **one dedicated** waiting line for all jobs.
- **Each** job is served by **any one** server/machine.



Introduction to Operations Management (Flow Process Diagram)

- n **serial/sequence** process, with **one dedicated** waiting line for all jobs.
- **Each** job has parts which are served individually by **all** servers/machines.



Introduction to Operations Management

(Key process metrics)

- **Inventory**: How much to store, when to order, what is the storage cost, etc.
- **Flow rate**: How much material moves in the process.
- **Flow time**: What is the average flow time in the process, are there any bottle necks.

Introduction to Operations Management

(Inventory = Flow Rate $\times$ Flow Time)

- A shipping company operates 10 container ships and each can carry 5,000 containers between ports.
- The company wants to load and unload 20,000 containers each week. Assume the ships always travel fully loaded.
- We need to find the longest average travel time (**Flow Time**) between ports that allows the company to meet their goal of 20,000 containers per week
- On average there are $10 \times 5000 = 50,000$ containers transported between ports, and the company wants to load/unload 20,000 containers each time.
- Hence **Flow Time** = $50000/20000 = 2.5$ weeks

Introduction to Operations Management

(Capacity = $\{m/\text{Processing Time}\}$)

- A call center has one operator ($m=1$) who answers incoming calls and it takes the operator 6 minutes on an average to answer one call.
- Hence the **Capacity** of the call center is = $(1/6)$ Call per Minute per operator = $(1/6) \times 60 = 10$ Calls per hour per operator
- If number of operators is $m=4$, hence **Capacity** of the call center is = $10 \times 4 = 40$ calls per hour in total, i.e., **Capacity** = $(4/6)$ Call per minute = $(4/6) \times 60 = 40$ Calls per hour

Introduction to Operations Management

(Flow Rate = Minimum {Demand, Process Capacity})

- Assume **Demand** is 10 units per 15 minutes while **Capacity** (ability to process) per machine is 30 units per 20 minutes.
- Thus, **Demand** is $10 \times 4 = 40$ units per $15 \times 4 = 60$ minutes (1 hour) while **Capacity** (ability to process) per machine is $30 \times 3 = 90$ units per $20 \times 3 = 60$ minutes (1 hour).
- Hence **Flow Rate or Throughput** = $\min(40, 90) = 40$ units per hour, and it is a **Demand Constrained** process.

Introduction to Operations Management

(Flow Rate = Minimum {Demand, Process Capacity})

- Assume **Demand** is 20 units per 10 minutes while **Capacity** per machine is 10 units per 30 minutes, and **there are 5 machines**.
- Thus **Demand** is $20 \times 6 = 120$ units per $10 \times 6 = 60$ minutes (1 hour) while **Capacity** (ability to process) per machine is $10 \times 2 = 20$ units per $30 \times 1 = 60$ minutes (1 hour).
- Hence **Flow Rate or Throughput** = $\min(120, 20 \times 5) = \min(120, 100) = 100$ units per hour, and it is a **Capacity Constrained** process.

Introduction to Operations Management

(**Utilization** = {**Flow Rate/Capacity**})

- For a production process the flow rate is 40 units per hour while capacity is 30 units per 30 minutes.
- Thus **Flow Rate or Throughput** is 40 units per 60 minutes (1 hour) while **Capacity** (ability to process) per machine is $30 \times 2 = 60$ units per $30 \times 2 = 60$ minutes (1 hour).
- Hence **Utilization** of the production process is $= 40/60 = 2/3$.

Introduction to Operations Management

(Cycle Time = {1/Flow Rate})

- For a departmental store the **Flow Rate or Throughput** is 275 customers per hour while **Capacity** is 300 customers per 1 hour **per** billing desk.
- The **Cycle Time** = $1/275 = 0.0037$ hour per customer = $0.0037 \times 60 \times 60$ (≈ 13) seconds per customer.
- With **Capacity** we have = $(60 \times 60)/300$ (= 12) seconds per customer.
- Departmental store is **able** to handle the **Flow Rate or Throughput** of 225 customers per hour.
- 1 Billing desk saves $(13 - 12) = 1$ seconds per customer per hour, hence a savings of $1 \times 225/60$ (= 3.75) minutes per hour.
- If departmental store is open from 1000 hours to 2000 hours, then a saving of 3.75×10 (= 37.5) minutes per day with 1 Billing desk.

Introduction to Operations Management

(Cycle Time = {1/Flow Rate})

- For a departmental store the **Flow Rate or Throughput** is 250 customers per hour while **Capacity** is 100 customers per 1 hour **per** billing desk.
- The **Cycle Time** = $1/250 = 0.004$ hour per customer = $0.004 \times 60 \times 60 (= 14.4)$ seconds per customer.
- With **Capacity** we have = $(60 \times 60)/100 (= 36)$ seconds per customer.
- Departmental store is **unable** to handle the **Flow Rate or Throughput** of 250 customers per hour.
- With **3** billing desk **Capacity** = $\{(60 \times 60)/(100 \times 3)\} (= 12)$ seconds per customer.
- **3** Billing desks save $(14.4 - 12.0) = 2.4$ seconds per customer, hence a savings of $2.4 \times 250/60 (= 10)$ minutes per hour.
- If departmental store is open from 1000 hours to 2000 hours, then a saving of $10 \times 10 (= 100)$ minutes per day with **3** Billing desks.

Introduction to Operations Management

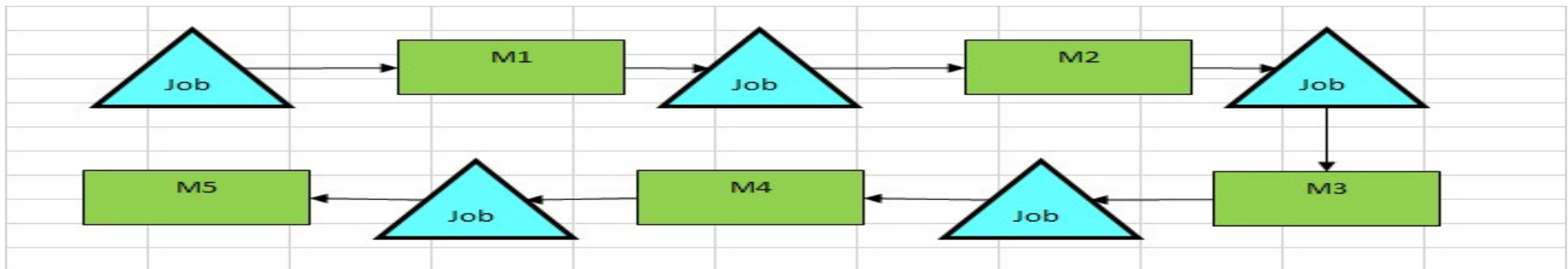
(Example # 02)

- A primary care hospital at Coimbatore (10.9974° N, 76.9589° E), has the **Capacity** to see 350 patients per day. The **Demand** rate is, however, only 270 patients per day.
- **Flow Rate** = $\text{Min}(270, 350) = 270$ patients per day
- **Utilization** = $270/350 = 77.15\%$
- **Cycle Time** = $(1/270)$ day per patient = $(1/270) \times 24 \times 60$ (≈ 5.33) minute per patient

Introduction to Operations Management

(Example # 03)

- There are 5 machines ($i=1,2,3,4,5$) in series in the shop floor at TATA Motors Lucknow (26.8467° N, 80.9462° E) Plant with average **processing times** of 3, 4, 5, 6 and 5 minutes per job respectively. While average **Demand** is 25 jobs per hour



Introduction to Operations Management

(Example # 03)

- **Capacity**($i=1$) = $(1/3) \times 60 = 20$ jobs per hour
- **Capacity**($i=2$) = $(1/4) \times 60 = 15$ jobs per hour
- **Capacity**($i=3$) = $(1/5) \times 60 = 12$ jobs per hour
- **Capacity**($i=4$) = $(1/6) \times 60 = 10$ jobs per hour
- **Capacity**($i=5$) = $(1/5) \times 60 = 12$ jobs per hour
- Processing **Capacity** = $\text{Min}(20, 15, 12, 10, 12) = 10$ jobs per hour
- **Flow Rate** = $\text{Min}(25, 10) = 10$ jobs per hour
- **Utilization**($i=1$) = $10/20 = 50.0\%$
- **Utilization**($i=2$) = $10/15 = 66.7\%$
- **Utilization**($i=3$) = $10/12 = 83.3\%$
- **Utilization**($i=4$) = $10/10 = 100.0\%$
- **Utilization**($i=5$) = $10/12 = 83.3\%$
- **Cycle Time** = $(1/10)$ hour per job = $(1/10) \times 60 = 6$ minutes per job

Introduction to Operations Management

(Example # 04)

- Consider a three-step airport security at IGIA, DEL (28.7041° N, 77.1025° E), where
 - ❖ **First step ($i=1$)**: Verifying ID and boarding pass and it takes 40 seconds per passenger.
 - ❖ **Second step ($i=2$)**: Security check and use of metal detector and it takes 20 seconds per passenger.
 - ❖ **Third step ($i=3$)**: Security check for hand luggage through an X-ray machine and it takes 75 seconds per passenger bag

Introduction to Operations Management

(Example # 04)

- Assume that there are many customers waiting in the process, then
 - ❖ Which resource is the bottleneck?
 - ❖ What is the capacity of the process?
 - ❖ What is the flow rate?
 - ❖ What is the utilization of the metal detector?
 - ❖ What is the cycle time?

Introduction to Operations Management

(Example # 04)

- Resource bottleneck: **Third step** ($i=3$).
- Capacity level of process: Capacity ($i=1$) = $1/40$; Capacity ($i=2$) = $1/20$; Capacity ($i=3$) = $1/75$, i.e., 1 passenger per 75 seconds.
- Flow rate: Flow Rate = Minimum {Demand, Process Capacity}, hence Flow Rate = $\text{Min}(\infty, 1/75) = 1$ passenger per 75 seconds which is 960 passenger per 20 hours (time the airport is open in a day).
- Utilization of the metal detector = Flow Rate/Capacity ($i=2$) = $(1/75)/(1/20) = 0.267$.
- Cycle time = $(1/\text{Flow Rate}) = (1/75) = 75$ seconds per passenger.

Introduction to Operations Management

(Example # 05)

- For the Subway restaurant in IIT Kanpur (26.4499° N, 80.3319° E) campus the average processing time to make a sandwich by a single-employee is 180 seconds/customer. DMS201 class ends at 10:30 AM (IST) and $m=30$ hungry students rush into Subway, each demanding a different sandwich. How long would it take to complete that demand? There are two cases, when all students arrive
 - ❖ Restaurant is empty with (i) $n=1$ employee serving and (ii) $n=4$ employees serving.
 - ❖ Restaurant already has 10 customers waiting to order with (i) $n=1$ employee serving and (ii) $n=4$ employees serving.

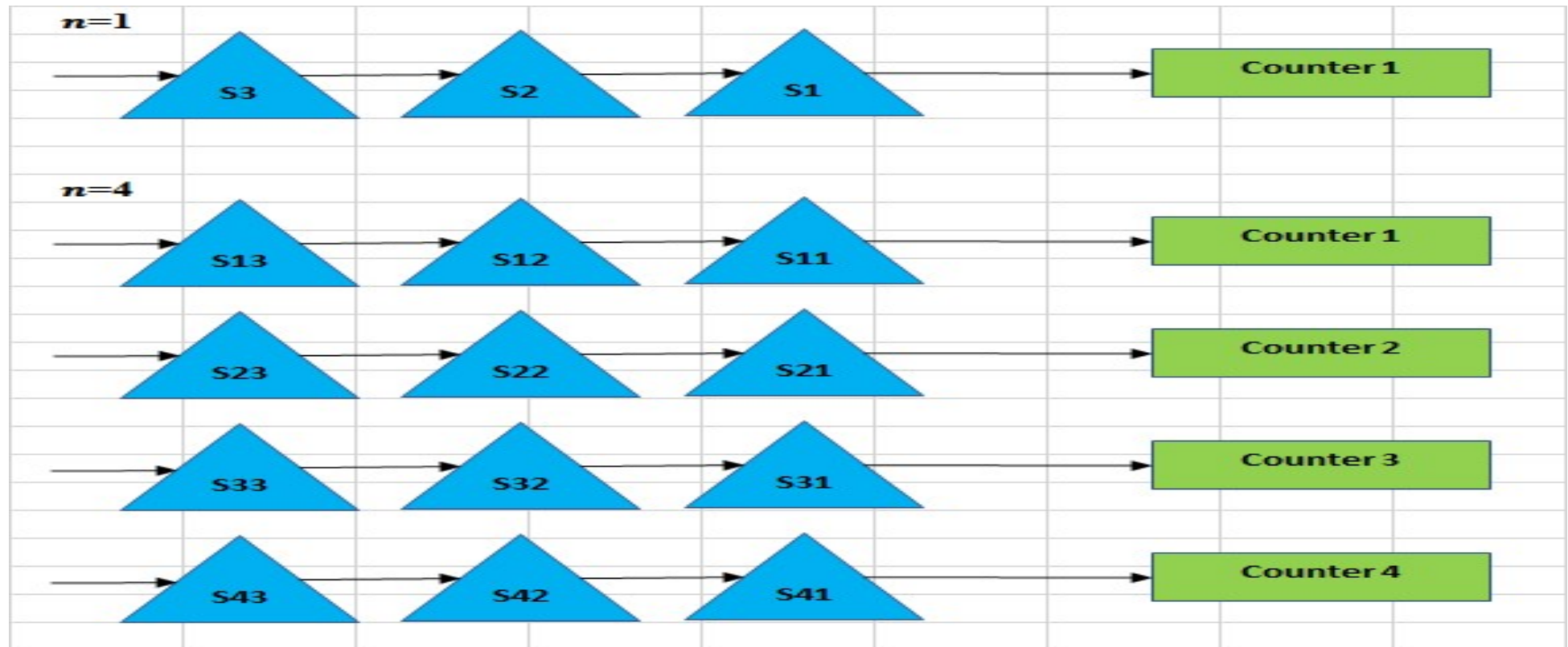
Introduction to Operations Management

(Example # 05)

- Flow Rate = $\text{Min}(\text{Demand}, \text{Capacity}) = \text{Min}(\infty, 1/180) = (1/180)$ customer per sec = $(1/180) \times 60 = (1/3)$ customer per minute
- Cycle Time = $(1/\text{Flow Rate}) = \{1/(1/180)\} = 180$ sec per customer per employee
- Time to process = $m \times (1/n) \times \text{Cycle Time} = 30 \times (1/1) \times 180 \text{ sec} = 1.5 \text{ hours per employee}$
- Time to process = $m \times (1/n) \times \text{Cycle Time} = 30 \times (1/4) \times 180 \text{ sec} = 22.5 \text{ minutes per employee}$

Introduction to Operations Management

(Example # 05)



Introduction to Operations Management

(Example # 05)

- Flow Rate = $\text{Min}(\text{Demand}, \text{Capacity}) = \text{Min}(\infty, 1/180) = (1/180)$
customer per sec = $(1/180) \times 60 = (1/3)$ customer per minute
- Cycle Time = $(1/\text{Flow Rate}) = \{1/(1/180)\} = 180$ sec per customer per employee
- Time to process = $10 \times (1/n) \times \text{Cycle Time} + m \times (1/n) \times \text{Cycle Time}$
 $= 10 \times (1/1) \times 180 \text{ sec} + 30 \times (1/1) \times 180 \text{ sec} = 2.0$ hours per employee with first DM201 student waiting time of 0.5 hours
- Time to process = $10 \times (1/n) \times \text{Cycle Time} + m \times (1/n) \times \text{Cycle Time}$
 $= 10 \times (1/4) \times 180 \text{ sec} + 30 \times (1/4) \times 180 \text{ sec} = 30$ minutes per employee with the first four DM201 students waiting time of 7.5 minutes

Introduction to Operations Management

(Example # 05)



Introduction to Operations Management

(Example # 06)

- Consider a three-step airport security at IGIA (DEL), (28.7041° N, 77.1025° E), where
 - ❖ **First step ($i=1$)**: Verifying ID and boarding pass and it takes 40 seconds per passenger.
 - ❖ **Second step ($i=2$)**: Security check and use of metal detector and it takes 20 seconds per passenger.
 - ❖ **Third step ($i=3$)**: Security check for hand luggage through an X-ray machine and it takes 75 seconds per passenger bag

Introduction to Operations Management

(Example # 06)

- In the morning at 6:30 M (IST) 25 passengers arrive, so how long will it take to serve these 25 passengers, and there are no passenger before.
- The process is empty, so we first have to compute the time until the first customer is served and we use the following.
- **Time to finish m units starting with an empty system = Time through an empty process + $[(m - 1) \times \text{Cycle time}]$.**

Introduction to Operations Management

(Example # 06)

- The first customer will take 40 seconds for verifying his/her ID and then 75 seconds at the X-ray machine, as the time of 20 seconds taken at the metal detector **does not matter** because it happens in parallel with the X-ray machine.
- Thus, it takes $(40 + 75)$ seconds = 115 seconds until the **first** customer is served.
- From then onward, we are serving a customer every 75 seconds, and as we have 25 customers to serve, this will take 25×75 seconds = 1875 seconds
- Total time taken $(115 + 1875)$ seconds = 33.2 minutes.

Introduction to Operations Management

(Important Processing Times)

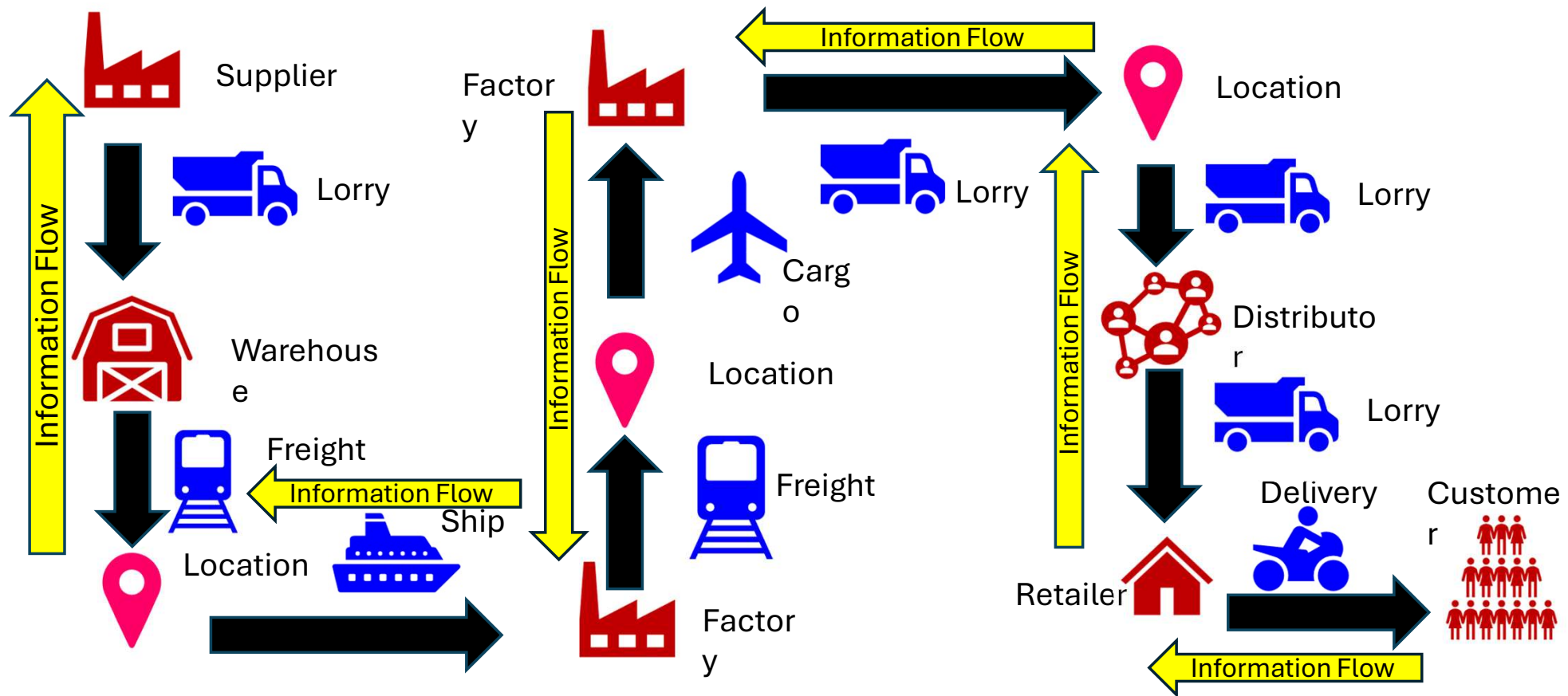
Processing time	Time to carry out all activities for a flow unit at one resource
Capacity of a resource	Number of workers at resource/processing time
Process capacity	Minimum of the capacity of all resources
Bottleneck	Resource with the smallest capacity
Flow Rate	Minimum{Demand, Process Capacity}
Utilization of a Resource	Flow rate/Capacity of the Resource
Utilization of a Process	Flow Rate/Process capacity
Cycle Time	1/Flow Rate
Time to produce m units (with a “full” system)	$m \times \text{Cycle Time}$
Time to produce m units starting with an empty process	Time through empty system + $[(m - 1) \times \text{Cycle Time}]$

Introduction to Operations Management (Supply Chain)

- Network of activities which facilitates delivery of raw/un-finished/finished product or service to warehouse/intermediaries/markets/the customer.
- Entities in this network are: Vendor/Supplier, Warehouse, Godown, Factory, Supplier, Retailer, Customer, Logistic/transport provider, Technology provider, etc.

Introduction to Operations Management

(Supply Chain Schematic Diagram)



Introduction to Operations Management

(Supply Chain Management (SCM))

- Is the vital business function that coordinates and manages all the activities of the supply chain linking suppliers, transporters, internal departments, third-party companies, and information systems.
- **Supply chain management (SCM)** for manufacturers entails: coordinating the movement of goods through the supply chain from suppliers to manufacturers to distributors to the final customers and sharing relevant information such as sales forecasts, sales data, etc., among members of the chain

Introduction to Operations Management

(**Bullwhip Effect in SCM**)

- Cascading variability effect due to inaccurate or distorted information as it travels along the supply chain system.
- Causes for **bullwhip effect** can be forecasting error, order batching, price fluctuation, demand fluctuation, shortage, etc.
- One can reduce **bullwhip effect** by proper coordination and information flow.

Introduction to Operations Management

(Bullwhip Calculation and Distribution of Order & Demand)

- **Bullwhip = (Variance of Order/Variance of Demand)**
 - Statistical Distribution for **Order**:
 - ❖ Slow-moving or discrete = Poisson or Negative Binomial.
 - ❖ Fast-moving, stable, high volume = Normal.
 - ❖ High variability = Gamma.
 - Statistical Distribution for **Demand**:
 - ❖ Low = Poisson.
 - ❖ High & stable = Normal.
 - ❖ High & lumpy = Negative Binomial.
 - ❖ Success/failure rate demand = Binomial.

Introduction to Operations Management (**Binomial Distribution**)

$$f(x) = {}^nC_x p^x q^{n-x}, x = 0, 1, 2, \dots, n$$

- n and p are the parameters where $p \in [0, 1]$, $q = (1 - p)$ and $n \in \mathbb{Z}^+$
- $E(X) = \mu_m = np$
- $V[X] = npq$
- Median = $\mu_e = \lfloor np \rfloor$ or $\lceil np \rceil$
- Mode = $\mu_o = \lfloor (n+1)p \rfloor$ or $\lceil (n+1)p \rceil - 1$

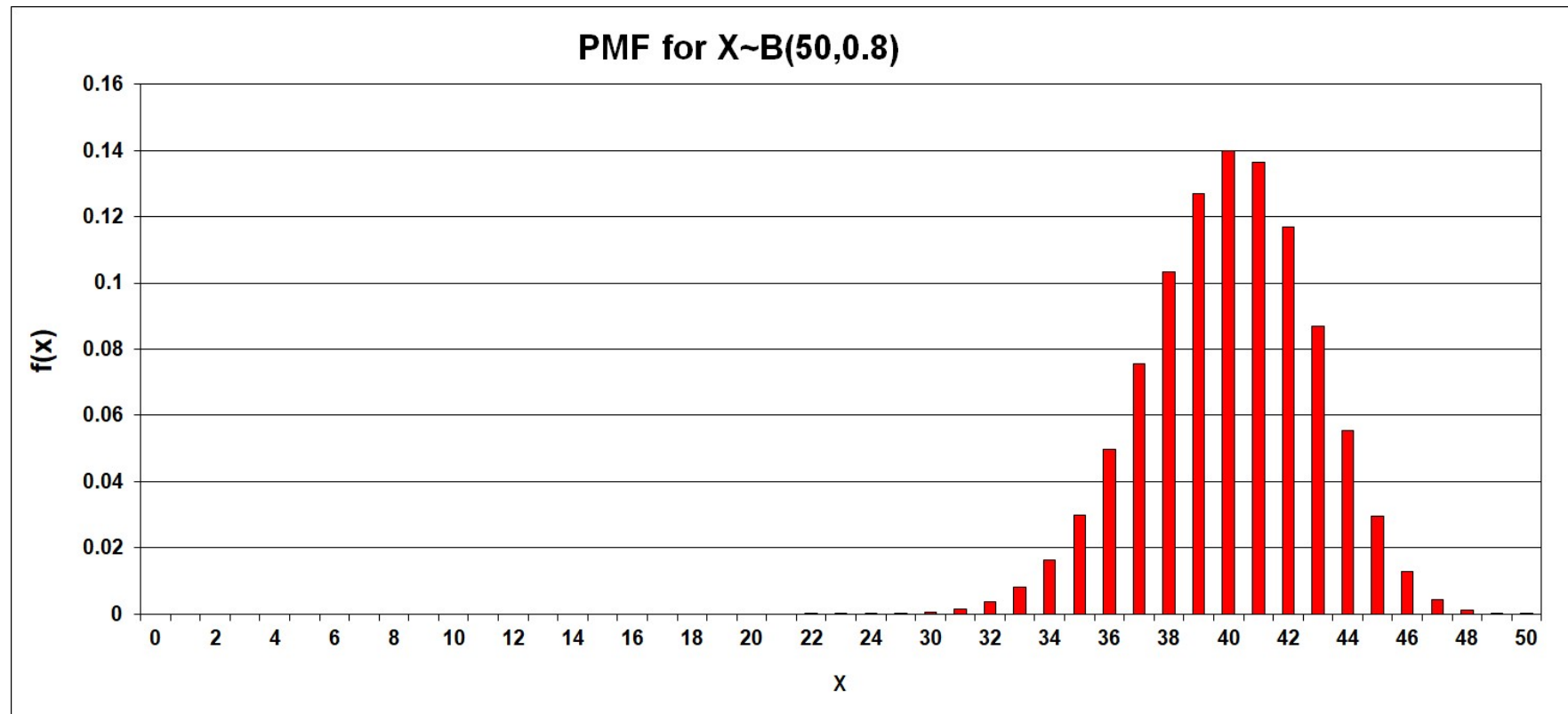
Introduction to Operations Management

(Binomial Distribution)

- Consider you are checking the quality of the product coming out of the shop floor.
- A product can either pass (with probability $p = 0.8$) or fail (with probability $q = 0.2$) and for checking you take such 50 products ($n = 50$).
- If X is the random variable denoting the **number of success** in these 50 inspections, we have $X \sim {}^{50}C_x (0.8)^x (0.2)^{50-x}$

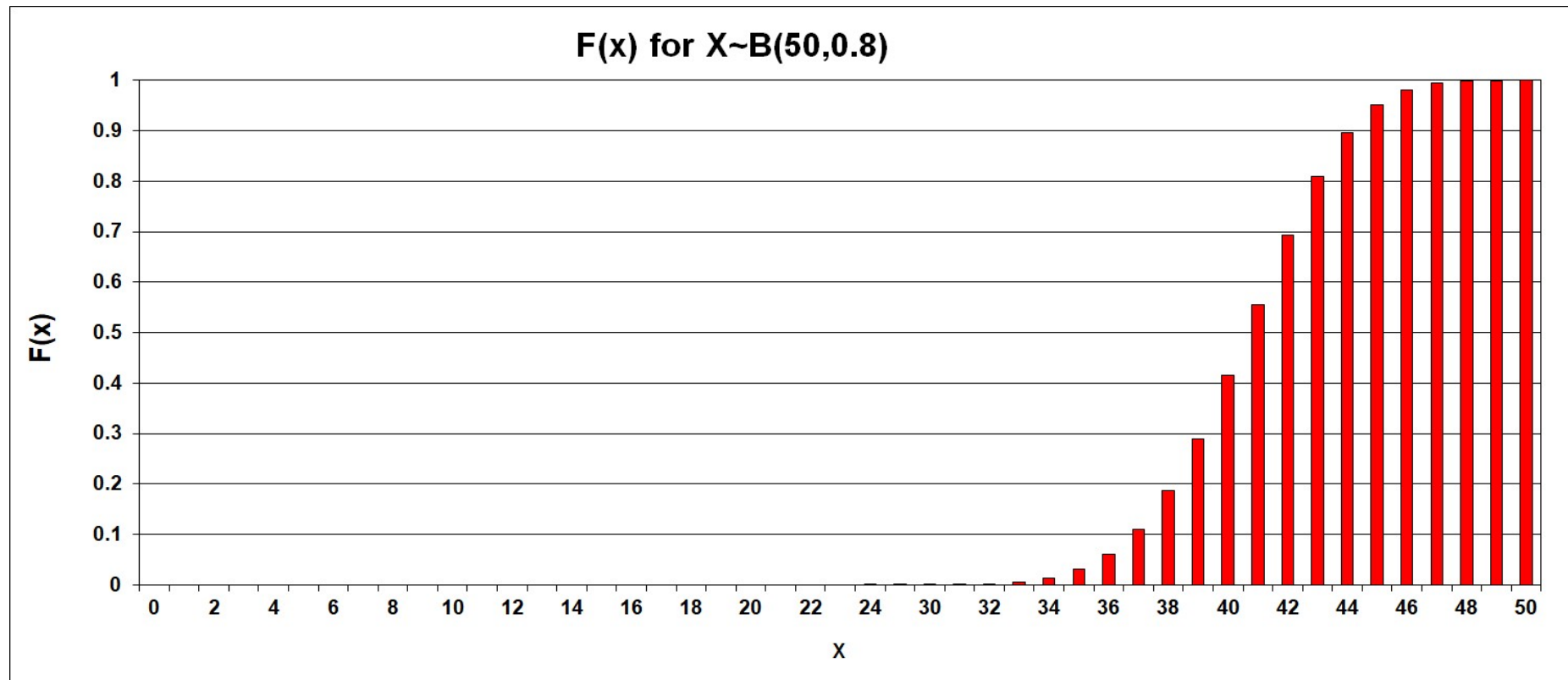
Introduction to Operations Management

(Binomial Distribution: $f(x)$)



Introduction to Operations Management

(Binomial Distribution: $F(x)$)



Introduction to Operations Management

(Binomial Distribution: Example # 07)

- 20% of products in the production line are defective and a random sample of 5 shares is chosen. Then what is the probability that at most 2 of them will be found to be defective?
- Here $p=0.2$, $q=0.8$.
- Hence the required probability is:
 - ❖ $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$
 - ❖ $P(X \leq 2) = {}^5C_0 0.2^0 0.8^5 + {}^5C_1 0.2^1 0.8^4 + {}^5C_2 (0.2)^2 (0.8)^3$
- $E(X) = 5 \times 0.2 = 1.0$
- $V(X) = 5 \times 0.2 \times 0.8 = 0.8$

Introduction to Operations Management

(Negative Binomial Distribution)

$$f(x) = {}^{r+x-1}C_{r-1} p^r q^x, x = 0, 1, \dots$$

- p and r are the parameters where $p \in [0, 1]$, $q = 1 - p$ and $r \in \mathbb{Z}^+$
- $E(X) = \mu_m = rq/p$
- $V[X] = rq/p^2$
- Median = μ_e = No single value
- Mode = $\mu_o = \lfloor (r-1)q/p \rfloor$

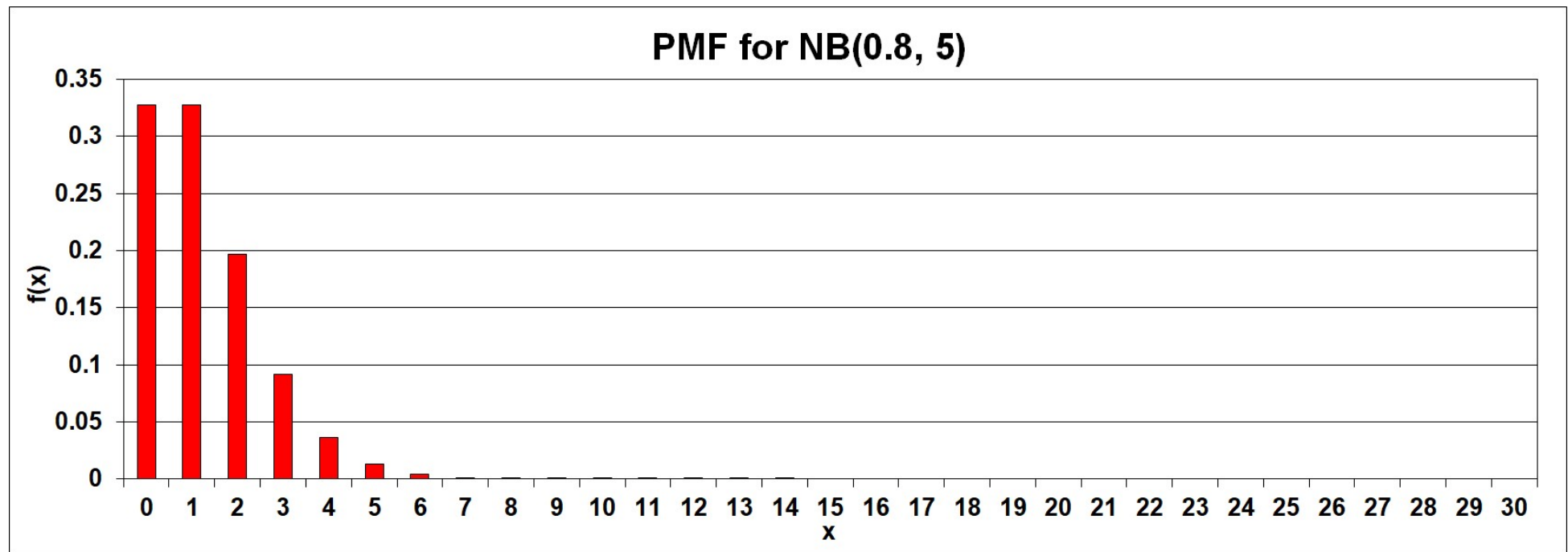
Introduction to Operations Management

(Negative Binomial Distribution)

- Consider the example where you are inspecting items from the production line.
- Now you are interested in finding the probability distribution of the **number of failures preceding the 5th success of getting the right product.**
- Then, if $p=0.8$, $q=0.2$, $X \sim {}^{5+x-1}C_{5-1}(0.8)^5(0.2)^x$

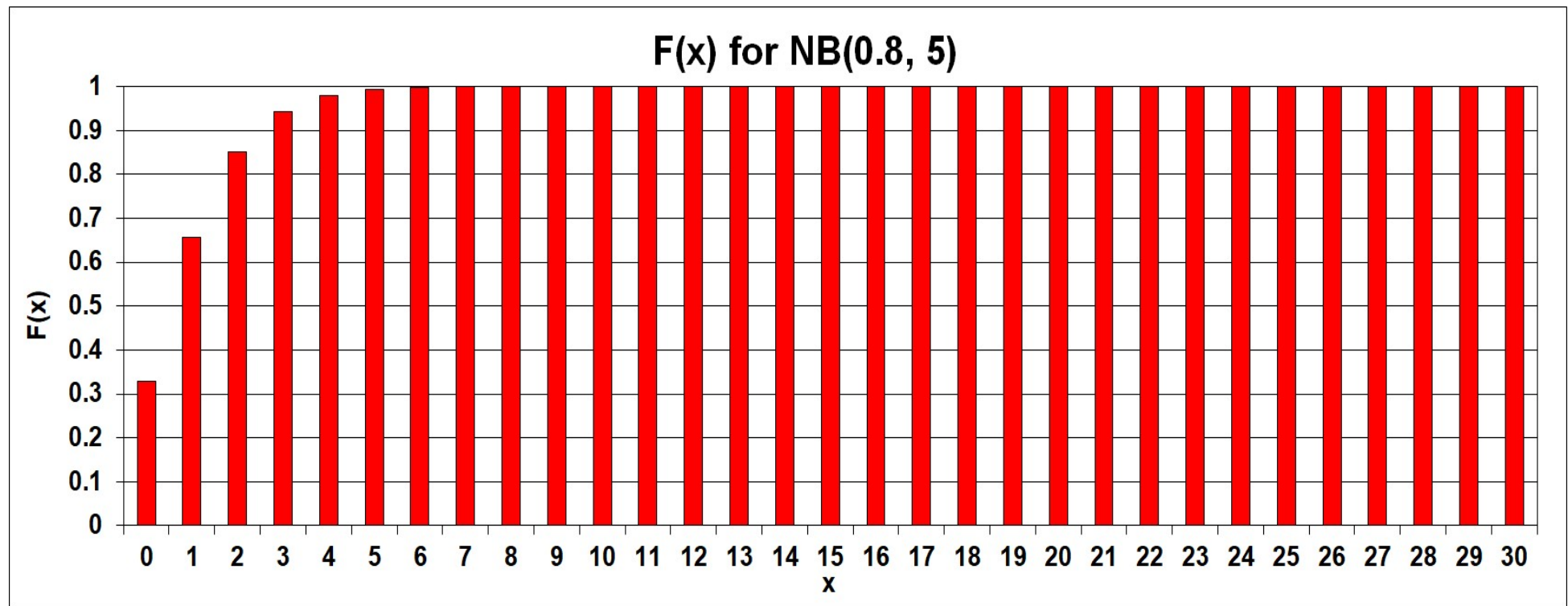
Introduction to Operations Management

(Negative Binomial Distribution: $f(x)$)



Introduction to Operations Management

(Negative Binomial Distribution: $F(x)$)



Introduction to Operations Management

(Negative Binomial Distribution: Example # 08)

- Suppose that the probability of a manufacturing process producing a defective item is 0.05 and assume that the quality of any one item is independent of the quality of any other item produced.
- If a Mr. Rao the quality control officer selects items at random from the production line and stops his inspection when he gets 10 good items, then what is the probability of getting 2 defective items before he stops inspection?
- Here $p=0.95$, $q=0.05$, hence required probability is $P(X=2) = {}^{10+2-1}C_{10-1} (0.95)^{10} (0.05)^2$
- $E(X) = (5 \times 0.05) / 0.95 = 0.263$
- $V(X) = (5 \times 0.05) / 0.95^2 = 0.277$

Introduction to Operations Management (Poisson Distribution)

$$f(x) = e^{-\lambda} \lambda^x / x!, \quad x = 0, 1, 2, \dots$$

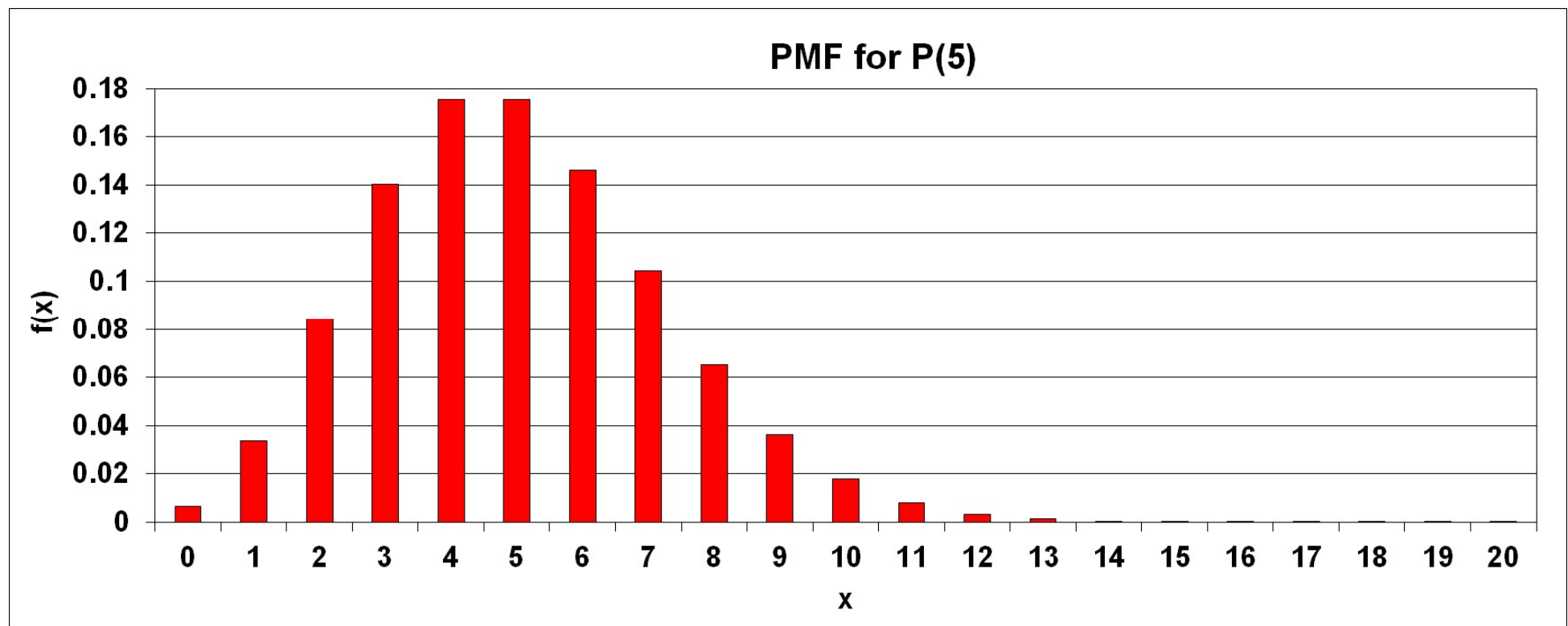
- λ is the parameter where $\lambda > 0$
- $E(X) = \mu_m = \lambda$
- $V[X] = \lambda$
- Median = $\mu_e = \lfloor \lambda + 1/3 - 1/(50\lambda) \rfloor$
- Mode = $\mu_o = \lceil \lambda \rceil - 1$ or $\lfloor \lambda \rfloor$

Introduction to Operations Management (Poisson Distribution)

- Consider the arrival of the number of customers at the bank teller counter.
- If we are interested in finding the probability distribution of the **number of customers** arriving at the counter in specific intervals of time and we know that the average number of customers arriving is 5.
- Then, we have $\lambda=5$, $X \sim e^{-5} 5^x / x!$

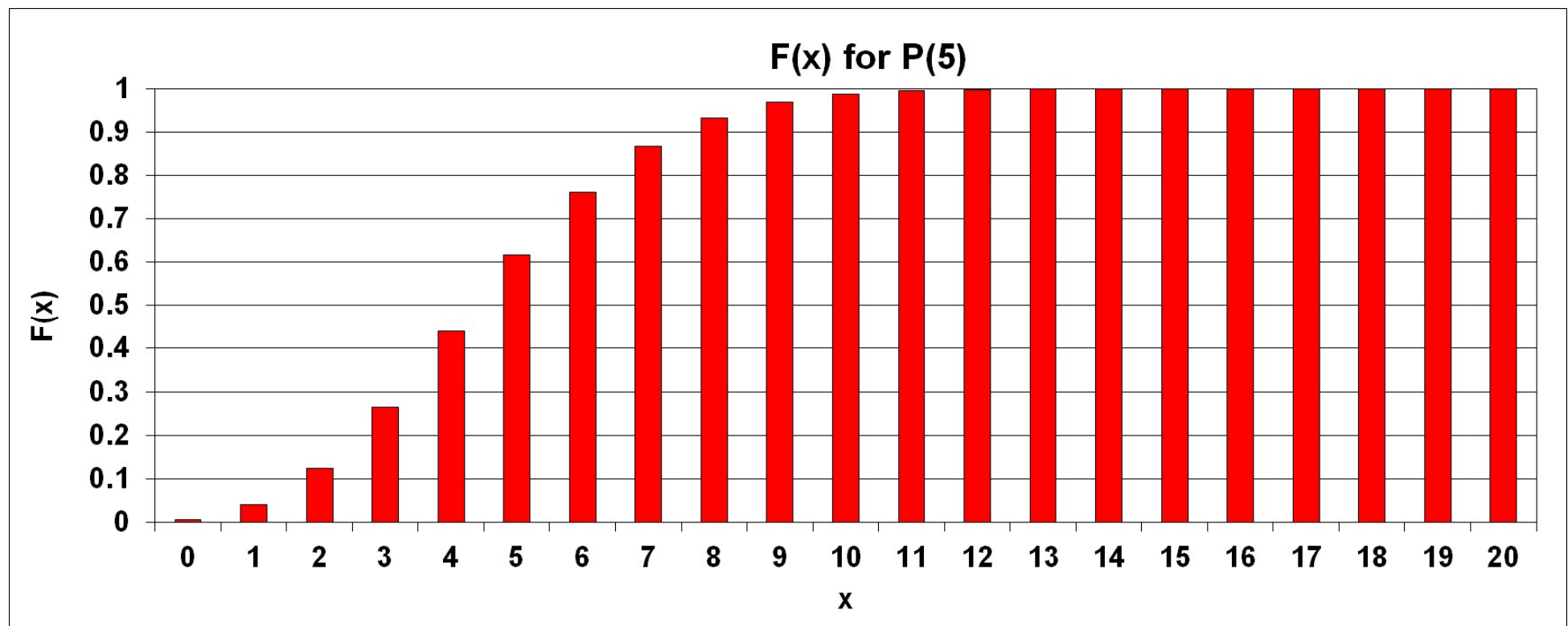
Introduction to Operations Management

(Poisson Distribution: $f(x)$)



Introduction to Operations Management

(Poisson Distribution: $F(x)$)



Introduction to Operations Management

(Poisson Distribution: Example # 09)

- Mr. Gurneek Singh who is the cashier at the departmental store cash counter notices that the average number of customer arriving at the cash counter per 5 minutes is 10.
- Then what is the probability that more than 5 customers arrive at the cash counter with the interval of 5 minutes?
- Here $\lambda=10$, hence the required probability is
 - ❖ $P(X \geq 6) = 1 - \{P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)\}$
 - ❖
$$P(X \geq 6) = 1 - \left\{ \frac{\exp(-10) \times 10^0}{0!} + \frac{\exp(-10) \times 10^1}{1!} + \frac{\exp(-10) \times 10^2}{2!} + \frac{\exp(-10) \times 10^3}{3!} + \frac{\exp(-10) \times 10^4}{4!} + \frac{\exp(-10) \times 10^5}{5!} \right\}$$

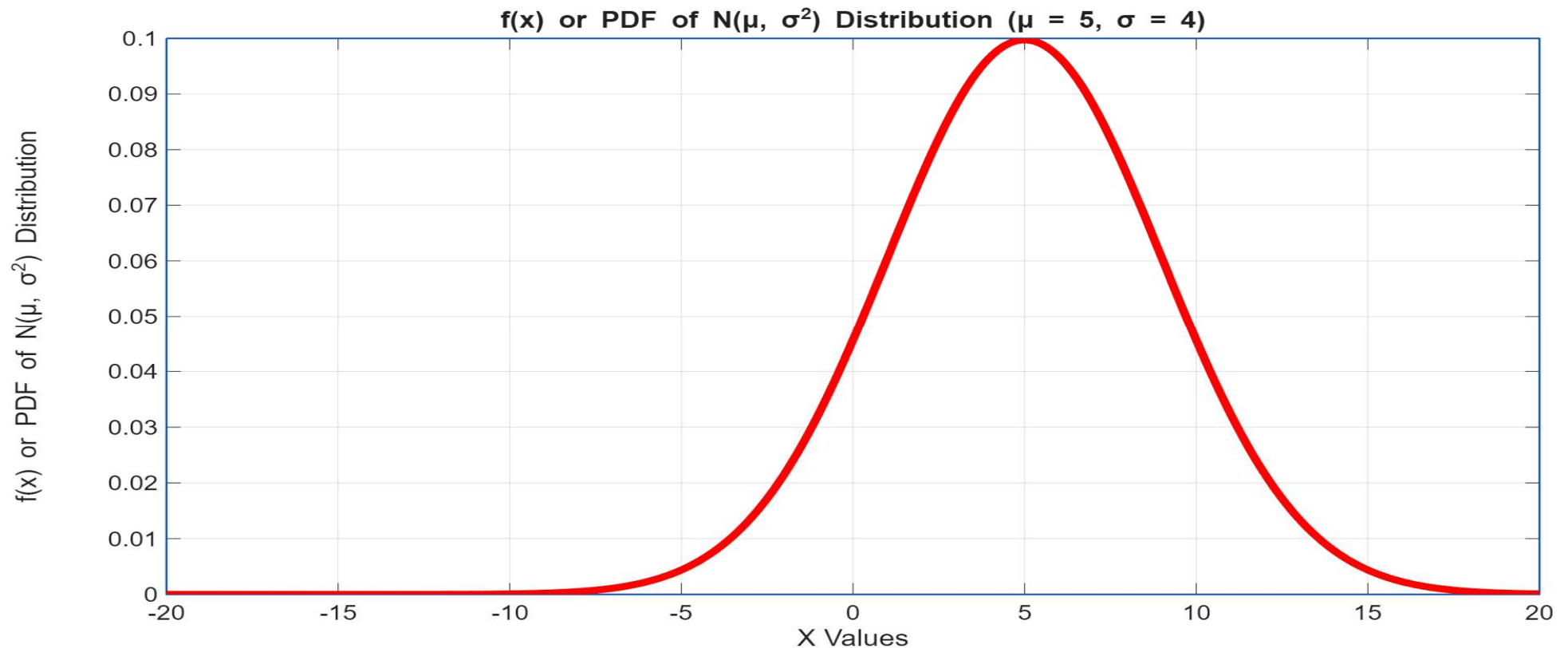
Introduction to Operations Management (Normal Distribution)

$$f(x) = \{1/(\sqrt{2\pi}\sigma)\}\exp[-(x - \mu)^2/2\sigma^2], -\infty < x < +\infty$$

- μ (location), σ^2 (σ : scale) are the parameters where $\mu \in \mathbb{R}$ and $\sigma^2 > 0$
- $E(X) = \mu_m = \mu$
- $V[X] = \sigma^2$
- Median = $\mu_e = \mu$
- Mode = $\mu_o = \mu$

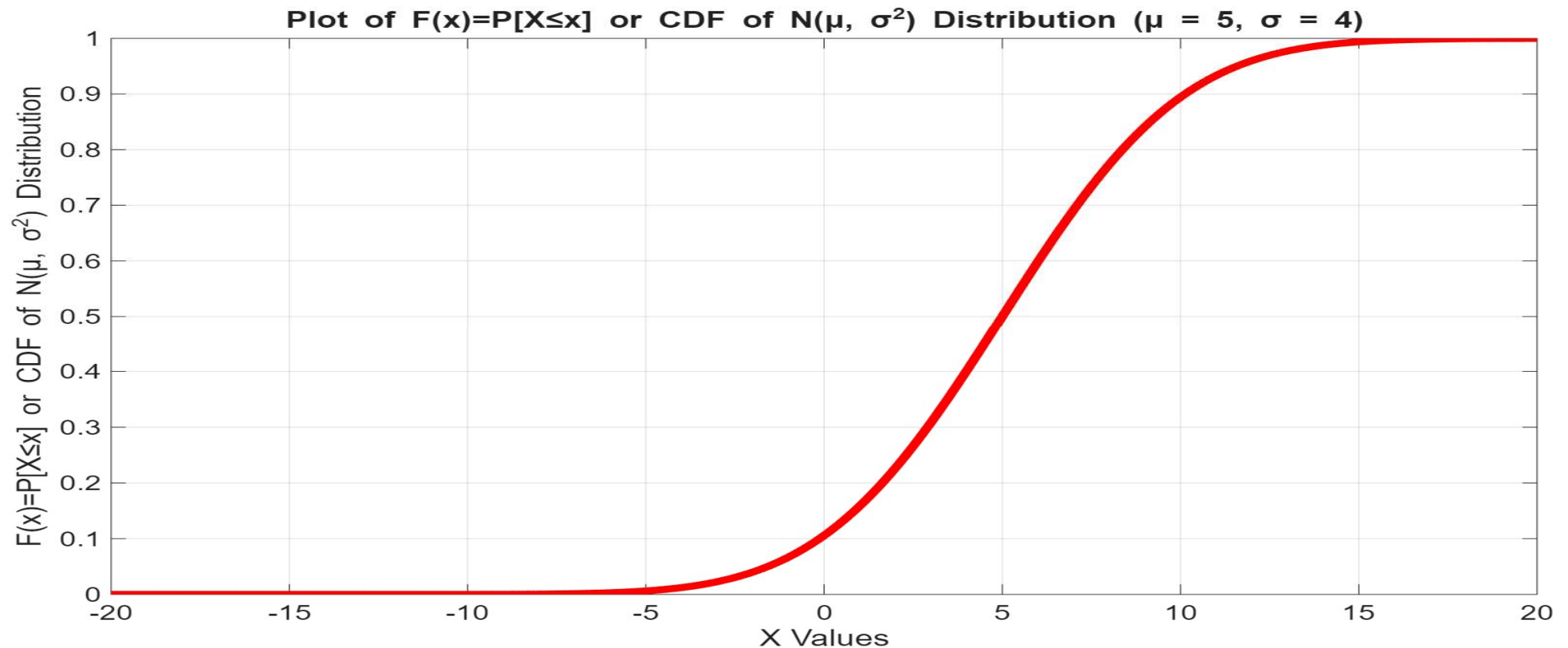
Introduction to Operations Management

(Normal Distribution: $f(x)$)



Introduction to Operations Management

(Normal Distribution: $F(x)$)



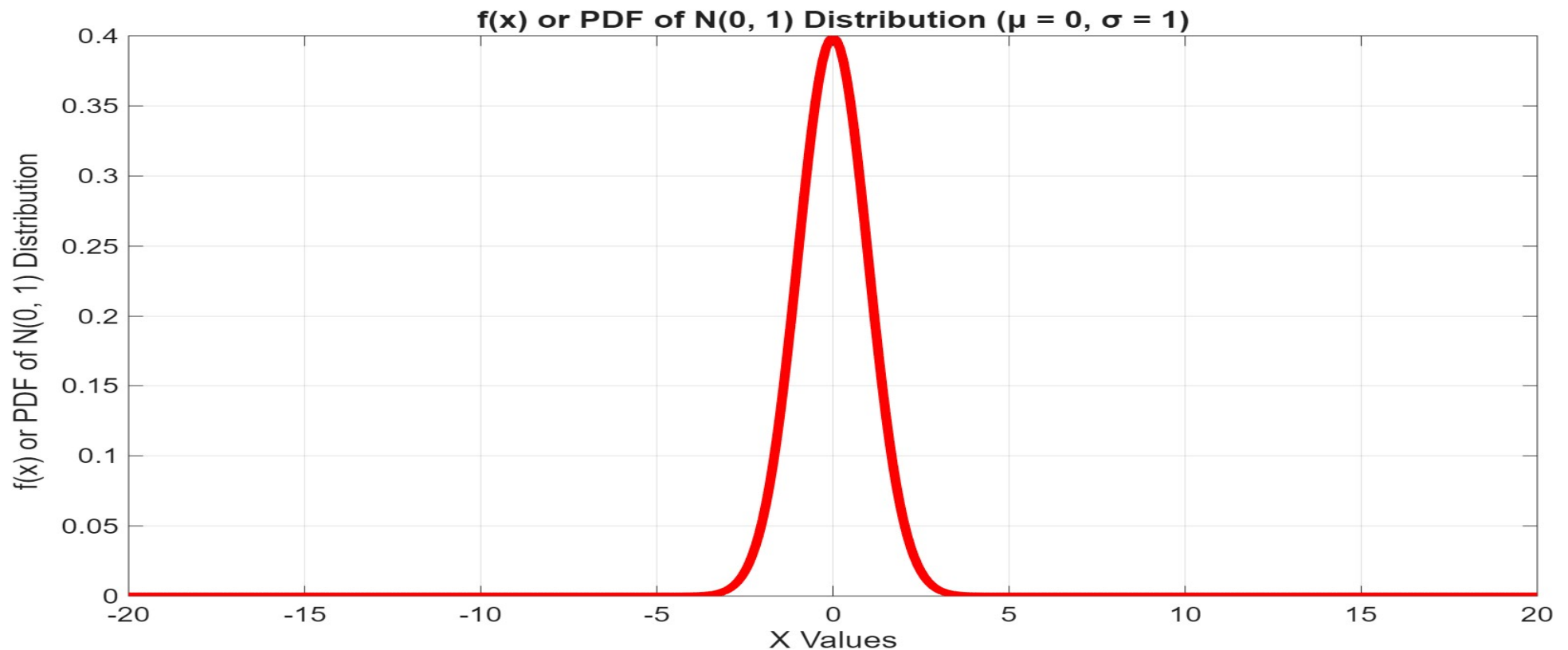
Introduction to Operations Management

(Standard Normal Distribution)

- Putting $Z=(X-\mu_x)/\sigma_x$ in the normal distribution with r.v., X , $X\sim N(\mu_x, \sigma_x^2)$ and we obtain the standard normal distribution of the form $f(z) = \{1/(\sqrt{2\pi})\}\exp(-z^2/2)$, where $\mu_z = 0$ and $\sigma_z^2 = 1$, i.e., $Z\sim N(0,1)$
- $F(x) = P(X \leq x) = F(z) = F(Z \leq z)$
- $f(z) = \phi(z)$
- $F(z) = F(Z \leq z) = \int_{-\infty < Z \leq z} f(z)dz = \int_{-\infty < Z \leq z} dF(z) = \Phi(z)$
- $F(Z \geq z) = 1 - F(Z \leq z) = 1 - \Phi(z)$

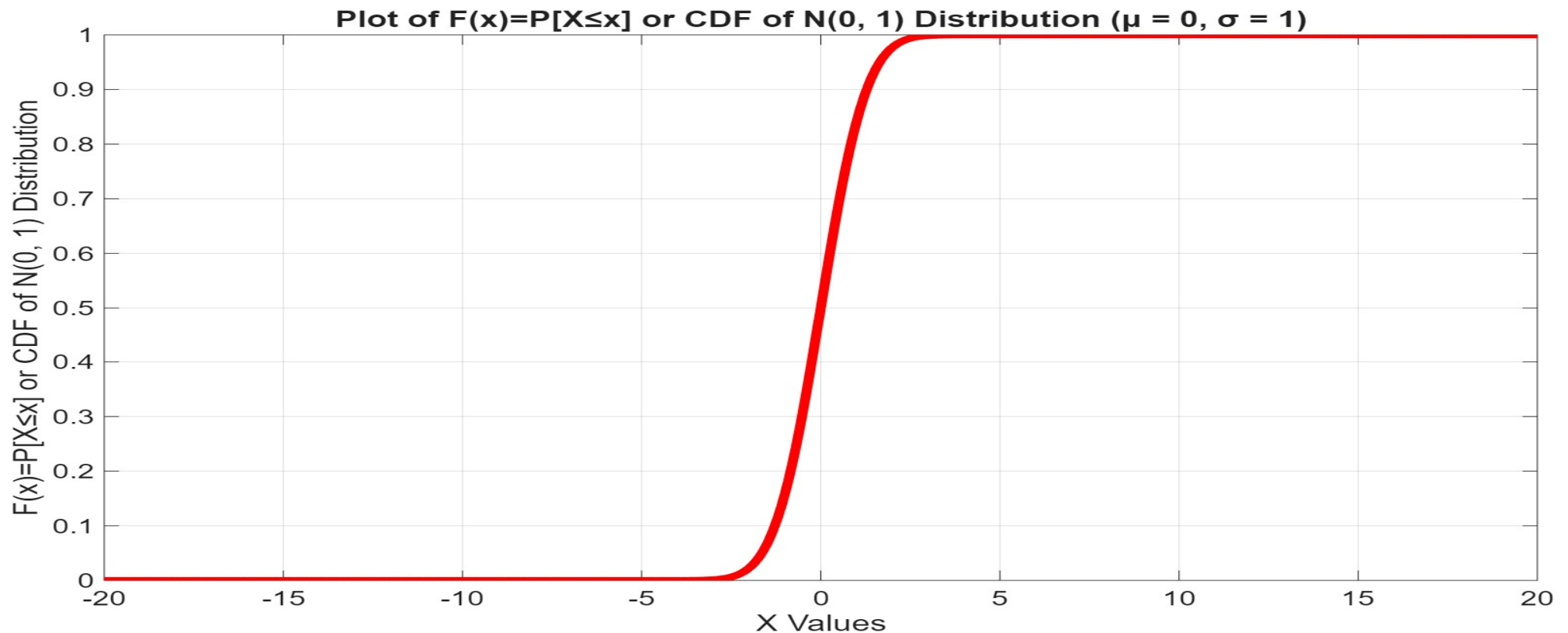
Introduction to Operations Management

(Standard Normal Distribution: $f(z)$)



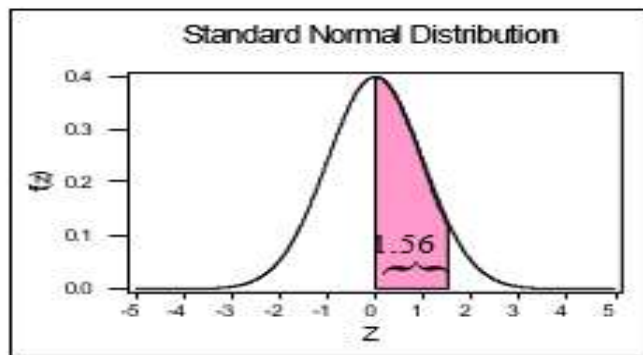
Introduction to Operations Management

(Standard Normal Distribution: $F(z)$)



Introduction to Operations Management

(Standard Normal Distribution)



Look in row labeled 1.5 and
column labeled .06 to find
 $P(0 \leq z \leq 1.56) = .4406$

Standard Normal Probabilities										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890

Introduction to Operations Management

(Normal Distribution: Example # 10)

- In an examination 20% of the students failed (i.e., obtained a score which is less than or equal to 40 marks out of 100) and 10% of the students obtained a grade A (score of 70 marks or above out of 100). Assuming normal distribution of marks find the mean and the standard deviation of the distribution.
- $P(X \leq 40) = 0.2 = P[(X - \mu_X)/\sigma_X \leq (40 - \mu_X)/\sigma_X] = P(Z \leq z_1) = \Phi(z_1) = -0.84$
- $P(X \geq 70) = 0.1 = P[(X - \mu_X)/\sigma_X \geq (70 - \mu_X)/\sigma_X] = P(Z \geq z_2) = 1 - P(Z \leq z_2) = 1 - \Phi(z_2), \Phi(z_2) = 0.9$
- Solve the two equations to obtain: $z_1 = (40 - \mu_X)/\sigma_X = -0.84$ and $z_2 = (70 - \mu_X)/\sigma_X = +0.90$
- $\mu_X = 54.12; \sigma_X = 17.64$

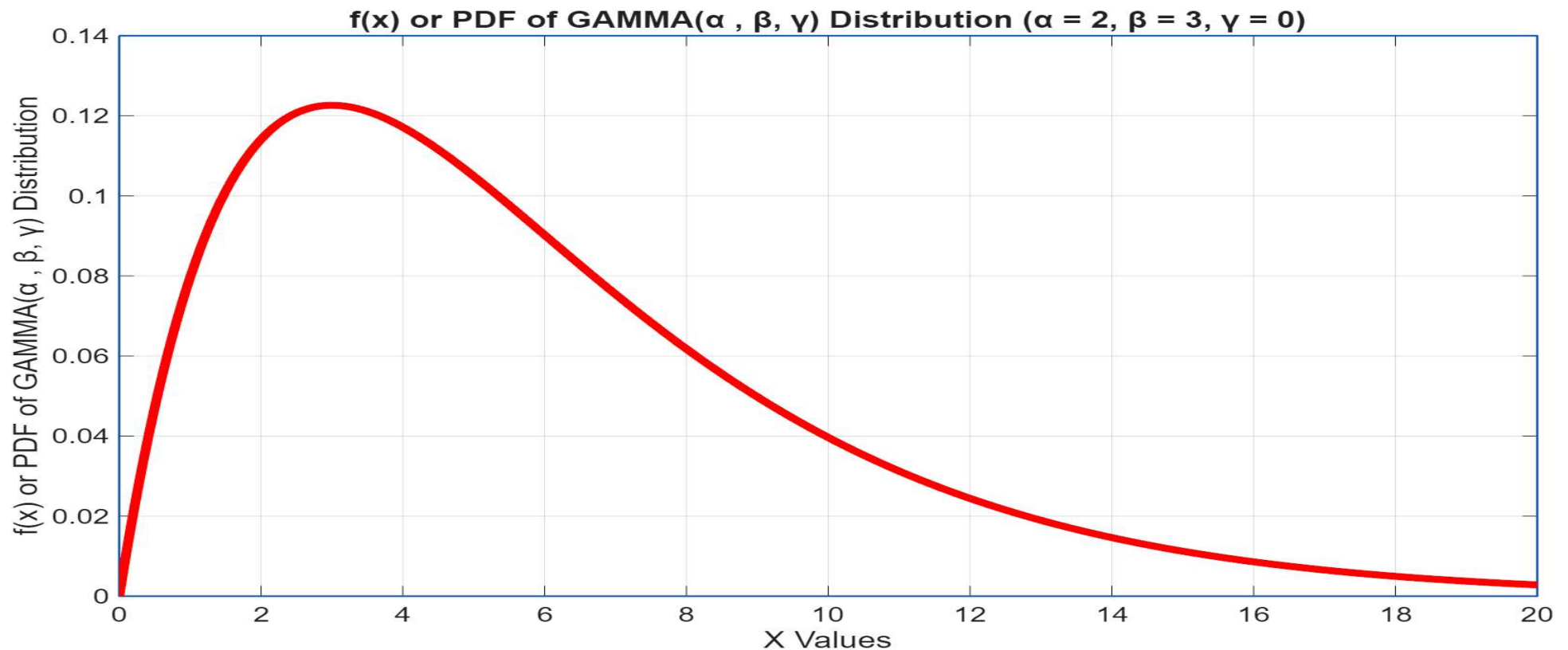
Introduction to Operations Management (Gamma Distribution)

$$f(x) = \beta^{-\alpha} \Gamma^{-1}(\alpha) (x-\gamma)^{(\alpha-1)} \exp\{-(x-\gamma)/\beta\}, x > \gamma$$

- $\alpha > 0$: Shape parameter
- $\beta > 0$: Scale parameter
- γ : Location parameter
- $E(X) = \mu_m = \alpha\beta + \gamma$
- $V[X] = \alpha\beta^2$
- Median = $\mu_e = \gamma + \beta \times x^*$, where x^* should be found from $P(X \leq x) = 1/\{\Gamma(\alpha)\beta^\alpha\} \int_0^x x^{\alpha-1} \exp(-x/\beta) dx = 1/2$
- Mode = $\mu_o = (\alpha - 1)\beta$

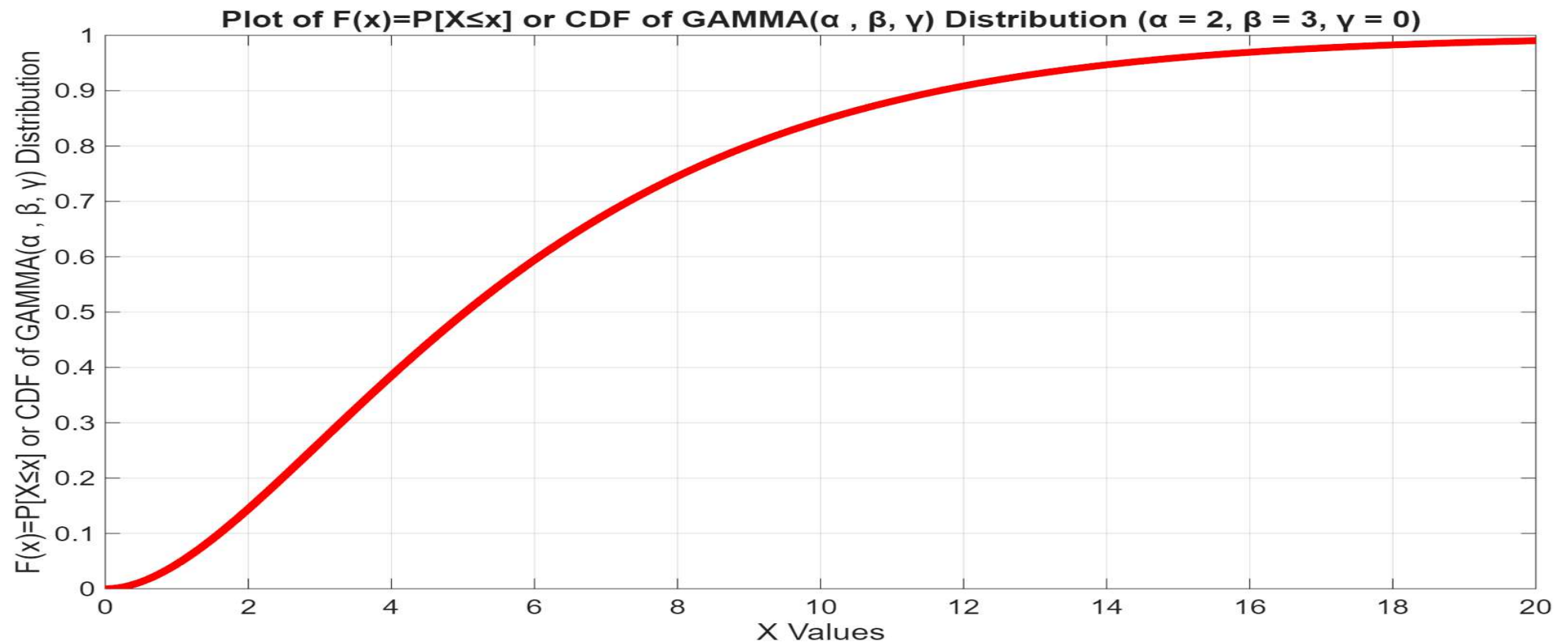
Introduction to Operations Management

(Gamma Distribution: $f(x)$)



Introduction to Operations Management

(Gamma Distribution: $F(x)$)



Introduction to Operations Management

(Gamma Distribution: Example # 11)

- Assume the time, Y , to fix a car is approximately a gamma distribution, with mean $\mu = 2$ hours and variance $\sigma^2 = 2$ hours², assume $\gamma = 0$.
- $E(X) = \mu_m = \alpha\beta + \gamma = \alpha\beta = 2$.
- $V[X] = \alpha\beta^2 = 2$
- Using the information we have, $\beta = 1$ and $\alpha = 2$

Introduction to Operations Management

(Bullwhip Value: Example # 12)

- **Order for slow moving items** is $X \sim P(\lambda=20)$, hence Variance = 20 and **Demand for low requirement items** is $X \sim P(\lambda=5)$, hence Variance = 5. Thus, Bullwhip value = $20/5 = 4$.
- **Order for slow moving items** is $X \sim NB(p=0.2, r=5)$, hence Variance = $5 \times 0.8 / 0.2^2 = 100$ and **Demand for low requirement items** is $X \sim P(\lambda=10)$, hence Variance = 10. Thus, Bullwhip value = $100/10 = 10.0$.
- **Order for high/stable volume items** is $X \sim N(\mu=12, \sigma^2=81)$, hence Variance = 25 and **Demand for high/stable volume items** is $X \sim N(\mu=20, \sigma^2=16)$, hence Variance = 16. Thus, Bullwhip value = $81/16 = 5.1$.
- **Order for high variability items** is $X \sim G(\alpha=2, \beta=3, \gamma=5)$, hence Variance = $2 \times 3^2 = 18$ and **Demand for high/lumpy volume items** is $X \sim NB(p=0.5, r=4)$, hence Variance = $4 \times 0.5 / 0.5^2 = 8$. Thus, Bullwhip value = $18/8 = 2.25$.